

Engagement That Sells: Influencer Video Advertising on TikTok

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Abstract. Many ads are engaging, but what makes them engaging may have little to do with the product. This problem can be particularly relevant to influencer advertising if influencers are motivated to promote themselves, not just the product. We develop an algorithm to measure the degree of effective engagement associated with the product and use it to predict the sales lift of influencer video advertising. We propose the concept of the product engagement score (PE score) to capture how engaging the product itself is as presented in a video. We estimate pixel-level engagement as a saliency map by training a deep three-dimensional convolutional neural network on video-level engagement data. We locate pixel-level product placement with an object detection algorithm. The PE score is computed as the pixel-level, engagement-weighted product placement in a video. We construct and validate the algorithm with influencer video ads on TikTok and product sales data on Taobao. We use variation in video posting time to identify video-specific sales lift and show that the PE score significantly and robustly predicts sales lift. We explore drivers of engagement and discuss how various stakeholders in influencer advertising can use the PE score in a scalable way to manage content, align incentives, and improve efficiency.

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Supplemental Material: The online appendix and instructions to acquire data are available at <https://doi.org/10.1287/mksc.2021.0107>.

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1. Introduction

“The next Amazon competitor is going to look like a social or video app, not a shopping app,” says Connie Chan, a general partner of the venture capital firm Andreessen Horowitz.¹ E-commerce is moving beyond utilitarian and search-driven platforms to embrace more entertaining and discovery-driven platforms. On the latter type of platforms, the lines between content and commerce are blurry. Content creators or influencers often engage with users and sell to them at the same time. This mixing of entertainment and commerce has given rise to the concept of “entertainment commerce.” One may even say that the letter *e* is now standing for *entertainment* in this emerging form of e-commerce.²

TikTok is one of the major platforms leading this transformation. Its core feature of short-form video sharing has attracted a massive following. As one of

the most downloaded apps in the world since 2018, TikTok had 1.5 billion monthly active users around the globe in 2023 and the number is expected to reach 1.8 billion by the end of 2024.³ E-commerce is well integrated into TikTok, especially in its original Chinese version. An ecosystem has been developed where product sellers routinely pay influencers to place products in their videos, whereas users make purchases using product links in the videos. According to a recent industry survey, TikTok is the most popular channel of influencer marketing by far, surpassing Instagram, YouTube, and Facebook.⁴

Despite its sharp rise, how influencer video advertising contributes to product sales is unclear. There is not yet a systematic way to predict an influencer video ad’s *sales lift*, meaning the incremental sales conversion attributed to the ad.⁵ As a result, sellers often rely on influencer engagement metrics (such as the

numbers of likes, comments, and shares) for campaign management. On TikTok, many sellers would simply choose an engaging influencer and then leave it to the discretion of the influencer to design a video ad. The result has been less than ideal. Anecdotes abound where influencer video ads are highly engaging but do a poor job of lifting sales.⁶

The goal of this paper is to develop a method to predict how much influencer video ads lift product sales. Our argument is that ads can be engaging for the “wrong” reason; what makes them engaging may have little to do with the advertised product. We call this ineffective engagement. Influencer advertising can be particularly susceptible to this problem because influencers are often incentivized to promote their personal brand, not just the product.⁷ As such, they may not want to allocate the most engaging space and time of their videos to the product, which lowers ad effectiveness. Based on this argument, we develop a metric for effective engagement, a metric that captures the extent to which engagement is driven by the product or the ad is engaging for the “right” reason. We call this metric *product engagement score* (PE score for brevity).

We operationalize the PE score so that it is intuitive, is able to turn unstructured video data into structured information, and is measurable prior to ad release for better campaign management. To meet these objectives, we define a video ad’s PE score as the average pixel-level engagement score over the space and time of a video in which a product is presented. We compute the PE score in three steps.

First, we construct a three-dimensional (3D) *engagement heat map* for each video to measure the importance of each pixel to overall video-level engagement. The three dimensions are the height and the width of each video frame in pixels and the length of the video in seconds. We train a deep 3D convolutional neural network (CNN) using video-level engagement data. A video’s engagement heat map is then derived as a pixel-level saliency map, which outputs the gradient of video-level engagement with respect to each pixel in the video.

Second, we construct a 3D *product heat map* for each video that has the same dimension as the engagement heat map. The product heat map shows whether the advertised product is present at a given pixel in a given frame of the video. We estimate the product heat map by matching an image of the product to each frame of the video with an object detection algorithm called “scale-invariant feature transform” (SIFT).

Third, we compute the PE score as the Frobenius inner product of the two 3D matrices, normalized by the total number of pixels of the video. The PE score can thus be interpreted as a video’s engagement level over the pixels in which a product is presented or

equivalently, how engaging the product is as presented in a video ad.

We hypothesize that a video ad with a higher PE score is more effective in lifting sales. Note that a video ad that is engaging overall or features the product throughout does not necessarily have a high PE score. In the former case, overall engagement may be driven by nonproduct content; in the latter case, product presentation may be uninteresting. A high PE score, as its name emphasizes, requires the product itself to be engagingly presented in a video ad.

We evaluate our method using a data set of influencer video ads on the original Chinese version of TikTok (referred to as TikTok for brevity hereafter) and their corresponding product sales revenue on Taobao from May to November 2019.⁸ Indeed, the data reveal no significant correlation between video engagement metrics and changes in sales. This observation echoes the industry’s criticism of engagement as an inadequate predictor of sales conversion in entertainment commerce. For a smell test of the incentive-misalignment argument, we also collect an auxiliary data set, in which influencers advertise their own products. Consistent with the argument, the PE score tends to be higher in these videos than in videos where influencers advertise for another party.

For the main test, we first estimate video-specific sales lift of influencer video ads via the difference-in-differences (DID) method using the variation in video posting time for causal identification. We then explore predictors of video-specific sales lift in both linear and nonlinear frameworks. Consistent with our hypothesis, the PE score is a significant and robust predictor of sales lift. Notably, overall video engagement and product-placement intensity do not predict sales lift. Being both engaging and intensive with product placement but doing so separately does not predict sales lift either. That is, simply making the video more engaging or featuring the product more may not help; it is product engagement as captured in the PE score that drives sales. Finally, influencer characteristics, such as popularity and the price they charge, do not predict sales lift, contrary to common belief.

We supplement the main analysis with an exploratory study of possible drivers of the PE score. Using the engagement heat map and a series of computer vision algorithms, we find that pixel-level engagement increases with human presence, sad or happy facial expressions, and stimulating or novel activities. Aligning product placement spatiotemporally with these elements of engagement may help enhance sales conversion.

The PE score can be practically valuable in several ways. First, we invested heavily in algorithm calibration such that the PE score can be easily computed for

a video ad in future applications, with the only data requirement being the video ad itself and a product image.⁹ This means that influencers can use the algorithm as an automated tool to test their videos in the creative process prior to release. Second, the PE score introduces a new contractual instrument to the influencer advertising space. Sellers can use the PE score to screen candidate videos or directly write a contract based on it. Influencers can use the PE score to signal their conversion efficacy beyond what engagement metrics are able to communicate. Platforms can design various policies to use the PE score for more accurate attribution and more efficient allocation. After all, the PE-score concept is built upon the two pillars of entertainment commerce—entertainment and commerce.

2. Related Research

The paper is inspired by and contributes to several streams of marketing research. First, we address a problem in influencer marketing (Avery and Israeli 2020). Rapidly growing over the last decade, influencer marketing is projected to become a \$24 billion industry by the end of 2024.¹⁰ It is a marketing strategy that uses the influence of key individuals or opinion leaders to drive consumers' brand awareness and purchase decisions (Brown and Hayes 2008). Social media is the main channel through which influencers influence. Social media influencers are first content creators and then marketers; they produce valuable content to cultivate and captivate a sizable number of followers and then, monetize their attention.

Research on influencer marketing has studied a range of topics, including influencer dissemination of ads (Gong et al. 2017), consumer trust (Lou and Yuan 2019), influencer versus celebrity endorsements (Schouten et al. 2020), influencer selection (Valsesia et al. 2020, Tian et al. 2024), cultural effect (Bentley et al. 2021), drivers of engagement (Leung et al. 2022), returns on influencer promotions (Huang and Morozov 2024), and the impact of influencer posts on copyrighted content (Li et al. 2024).

We contribute by studying one of the latest forms of influencer marketing—influencer video advertising. With the rapid growth of video as a communication tool, influencer video advertising is gaining popularity in practice and attention in academia. One recent paper particularly related to our work is Rajaram and Manchanda (2024). The authors developed an interpretable deep learning framework to study the impact of various video-ad modalities on engagement. We, too, analyze influencer video ad content, but we focus on sales conversion; we develop an algorithm to predict sales lift from ad content.

Our study of sales lift adds to the discussion of ad engagement versus conversion. A series of papers has

found that engagement does not guarantee conversion. Ad viewing may even be negatively related to buyer intent (Teixeira et al. 2014, Tucker 2015); virality does not always add value to the brand (Akpınar and Berger 2017); Facebook liking has no positive impact on consumer attitudes or purchases (John et al. 2017); and engaging contents, such as consumer selfies, may not improve purchase intentions (Hartmann et al. 2021). Using actual sales data, we also find that engagement does not guarantee conversion. Furthermore, we propose and validate a novel metric that connects engagement with conversion. We find that what drives conversion is not engagement per se but effective engagement associated with the advertised product.

This paper is also related to the marketing literature on video content design.¹¹ An established stream of research links video content with real-time viewer behaviors during the process of video consumption. Various measurement innovations have been developed, including handheld devices (Polsfuss and Hess 1991), “feeling monitor” computers (Baumgartner et al. 1997), eye tracking (Wedel and Pieters 2008, Teixeira et al. 2010), electroencephalography (Barnett and Cerf 2017), facial expression tracking (Liu et al. 2018), functional magnetic resonance imaging (Tong et al. 2020), and viewer live comments analysis (Zhang et al. 2020).

We contribute to the video content design literature along four dimensions. First, the literature has focused on movies or standard video ads. We study a new type of content—video ads produced by influencers. Influencer video ads can be fundamentally different from traditional video ads. In particular, influencers' incentives to promote themselves may affect ad content design. Second, many methods proposed in this literature rely on real-time viewer behavior data in response to new videos to forecast their market outcomes. We instead use historical observational data on video-level engagement to infer pixel-level engagement without directly measuring them. This means that our algorithm can be applied directly and in a scalable way to new videos prior to release.¹² Third, much of the literature has focused on time-series data to capture temporal video features. We made a nontrivial investment to extend the analysis to the pixel-moment level. This more-granular, spatiotemporal approach to video content design helps reveal further insights. Fourth, the literature has typically used predefined features to represent video content, whereas we take a data-driven approach without relying on handcrafted features—and we do so without sacrificing the interpretability of the PE-score algorithm. We turn to the algorithm, its theoretical motivation, and its computational development next.

3. Algorithm Construction

The PE-score concept is motivated by the distinctive shopping process on entertainment commerce platforms.

Users typically come to these platforms for entertainment. On TikTok, for instance, users often passively browse a stream of video feeds without a clear goal of searching for or purchasing a product.¹³ However, purchase interest can be activated in the process of consuming a video ad if the advertised product happens to grab user attention. Based on this idea, our hypothesis behind the PE score is that with other things being equal, the more engaging an advertised product is in an influencer video ad, the more effective the video ad will be in lifting sales.¹⁴ To operationalize this idea, we propose a three-step algorithm (where the first two steps can be done in reverse order).

1. Compute a pixel-level engagement heat map over the video ad to identify the most engaging spots of the video.

2. Compute a pixel-level product heat map over the video ad to identify when and where the product is featured in the video.

3. Compute the PE score as the normalized Frobenius inner product of the two heat maps to capture the average product engagement of the video.

We explain these three steps in detail in the following sections.

3.1. Engagement Heat Map

For each video ad, we first estimate an engagement heat map, which is a 3D matrix that captures the spatiotemporal variation of content engagement in the video. The three dimensions of the engagement heat map are the height and the width of each video frame in pixels and the length of the video in seconds. Specifically, we train a deep 3D CNN on historical video-level engagement data, based on which we extract a saliency map over the input video.

The CNN architecture is suitable for our problem because it is well known to be particularly good at image recognition (see Malik and Singh 2019 for a tutorial). We take a transfer learning approach by first extracting features from video frames with a CNN pretrained on ImageNet, namely, Xception (Chollet 2017), with the top classification layer removed¹⁵ and then feeding the feature sequence into a 3D convolution layer, which accounts for the temporal dependencies across frames (e.g., Tran et al. 2015).

We take the transfer learning approach for two reasons. First, the pretrained network is optimized for its performance on image recognition, which is directly relevant to our task. Transferring the knowledge encoded in this pretrained network to our context is computationally efficient. Second, building on a pretrained network reduces the number of parameters to be estimated and mitigates the risk of overfitting.

For the main analysis, we use each video's number of shares as the measure of engagement. Shares can be a stronger signal of engagement than likes and

comments as users are willing to endorse shared videos on their social networks.¹⁶ Shares are also a common subject of academic research (e.g., Akpınar and Berger 2017, Tellis et al. 2019). However, the paper's results are robust if we use likes or comments to measure engagement (see Online Appendix H.1).

Prior to training, we regress video-level raw engagement data on influencer fixed effects, product fixed effects, acoustic features, and transcript embeddings.¹⁷ We retain the residuals from the regression and use them as labels to train the 3D CNN. Using engagement residuals instead of raw engagement allows us to not only control for outliers but also, focus on the variation in engagement that is driven by the visual component of the video ad while holding other features that may affect engagement fixed.¹⁸ In the rest of the paper, video-level engagement refers to this "residualized" engagement value unless otherwise noted.

We focus on videos with spoken words so that a valid transcript can be extracted for training, although the paper's results are robust if we relax this requirement. As we will detail in Section 4, the sample we rely on to construct the algorithm contains 16,951 video ads. We train the 3D CNN on 10,000 videos, validate it on 3,500 videos, and test it on 3,451 holdout videos, all randomly chosen.

To appreciate the magnitude of the raw data for pixel-level analysis, consider a typical TikTok video. It is most commonly 15–60 seconds in length and has up to 60 frames per second (FPS). Each frame of standard resolution on TikTok contains $1,080 \times 1,920$ pixels. Finally, each pixel has three red, green, and blue (RGB) color channels. As a result, one 15-second, 60-FPS TikTok video would contain $15 \times 60 \times 1,080 \times 1,920 \times 3 = 5,598,720,000$ pixel values. To make the training process tractable, we subsample videos to one frame per second and resize each frame to a dimension of 224×224 pixels.¹⁹ This allows each video to be represented as a much more feasible $(S, 224, 224, 3)$ numerical array, where S is the duration of a video in seconds.

In the end, our full 3D CNN has over 7 million trainable parameters, has over 2 million input variables (with each pixel value in a given color channel being an input variable), and takes in over 20 billion data points (pixel values) in the training process.²⁰ We train the 3D CNN on a high-performance computing cluster using TensorFlow.²¹ It achieves an accuracy (one minus the mean absolute percentage error) of 73% on the test set. See Online Appendix A for more details on the network structure and the training process.

After training the 3D CNN, we use it to extract saliency maps on videos held out for downstream analysis (videos in the "sales panel" to be explained in Section 4). A saliency map (Simonyan et al. 2014) is a heat map over an original image that represents the gradient of the outcome variable with respect to this

image. The value at each pixel on a saliency map corresponds to the partial derivative of the outcome variable with respect to that pixel while holding other pixel values fixed. For images with color, we follow the common practice to compute three partial derivatives for each of its RGB color channels at a given pixel and take the maximum of the absolute values of the three derivatives as the saliency value at that pixel.²² We focus on the absolute value or magnitude of the derivatives per the standard implementation of saliency maps (e.g., Simonyan et al. 2014). This magnitude tells us how much the outcome variable, video-level engagement, changes with respect to changes in pixels of the input image or which pixels need to be changed to affect video-level engagement. Note that changing a pixel in a certain way may either increase or decrease video-level engagement. But, the fact that such changes matter means that users are probably paying attention to that pixel. We thus interpret a high absolute value of the derivative as high engagement or attention at that particular pixel. These inferred pixel-level engagement values in saliency maps have been shown to accurately predict the actual gaze map based on eye tracking data (Borji et al. 2013, Dupont et al. 2016).

We adapt the saliency map to videos, which are sequences of image frames. Importantly, instead of computing the gradient with respect to pixels frame by frame, we do so with respect to pixels in the entire video. This allows us to capture any dependency across video frames when deciding which pixels are driving engagement. We estimate saliency maps using the trained 3D CNN with tf-keras-vis.²³ See Online Appendix B for more details on the saliency map.

To understand the engagement heat map from an econometric perspective, we use one video-level engagement measure (the number of shares) to back out engagement scores (the partial derivatives) at each pixel in the video. This is analogous to a regression of video-level engagement on all pixel values in the video, except that our algorithm can handle high-dimensional inputs and arbitrary correlation between pixel values across space and time.²⁴ To be able to interpret pixel-level engagement as causal effects, we also need the “no design endogeneity” assumption, meaning that there are no omitted factors driving both pixel values and video-level engagement. It would be problematic if, for example, a video is engaging because a cheerful influencer uses bright pixels and engages the audience with a bright personality. The use of residualized video-level engagement mitigates this concern by controlling for influencer and product fixed effects as well as acoustics and spoken content.

3.2. Product Heat Map

For each video, we estimate a product heat map, which is a 3D matrix of the same dimension as the

engagement heat map. We do so by matching an advertised product’s image to each frame of a video to estimate when and where the product is placed.

We use the SIFT algorithm (Lowe 1999) for product detection.²⁵ A popular object detection tool, SIFT matches features across different images to identify the presence of an object in a cluttered scene. The key challenge is to make sure the key features of an object are robust to changes in scale, rotation, illumination, and viewpoint. The solution is intuitive. First, the “key points” of both the reference or query image (product) and the target image (video frame) are extracted; these key points are invariant to rotation and rescaling of the image. Then, the key points are matched between the reference and target images based on the distance of their characteristics, called key-point descriptors.²⁶

Because SIFT matches at the pixel level, the identified product pixels can be scattered in a video frame and do not necessarily enclose the entire product. We connect these pixels to create a convex hull and consider all pixels within the convex hull as product pixels. The resulting product heat map is a 3D matrix of binary values, where one indicates product presence at a pixel and zero indicates absence. See Online Appendix C for more details on the product heat map.

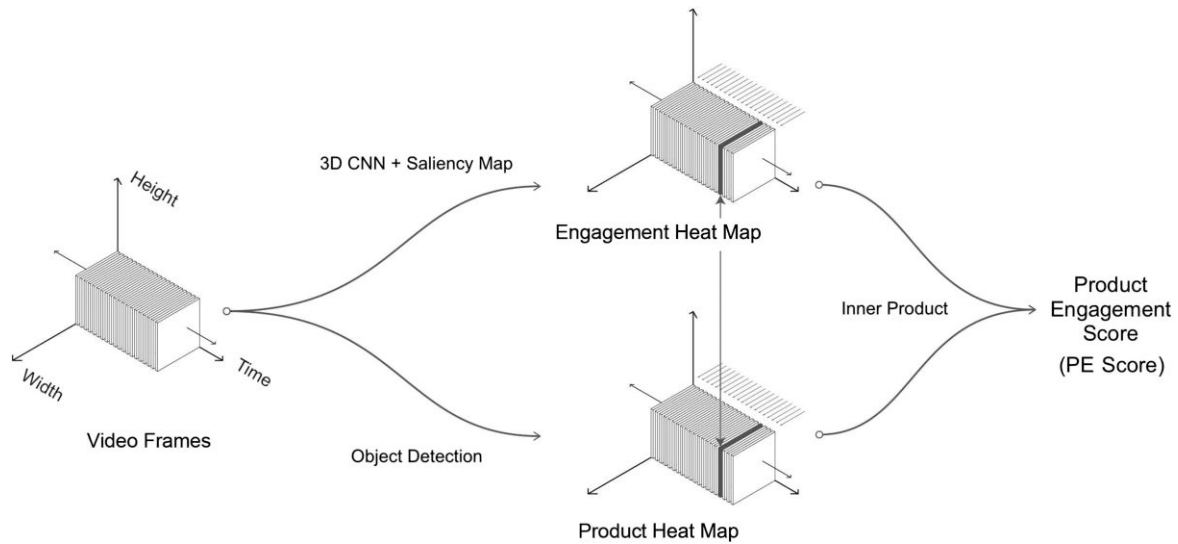
3.3. PE Score

In the third step, we combine the engagement and product heat maps to compute the PE score. Let e_{hwsv} be the (continuous) pixel-level value in the 3D engagement heat map and p_{hwsv} be the (binary) pixel-level value in the 3D product heat map. The symbols h , w , s , and v index height (in pixels), width (in pixels), time (in seconds), and video, respectively. We define a video’s PE score as the normalized Frobenius inner product of the two heat maps:

$$PE\ score_v := \frac{1}{H_v W_v S_v} \sum_{h,w,s} e_{hwsv} \cdot p_{hwsv}, \quad (1)$$

where H_v , W_v , and S_v are the height, width, and length of video v , respectively. As explained, we reshape the frame size for each video so that H_v and W_v are the same across videos, whereas we allow S_v to be different across videos. Their product, $H_v W_v S_v$, is the total number of pixels or the volume of video v . As discussed, we interpret the PE score as the average product engagement of a video; the inner product captures the spatiotemporal synchronicity between content engagement and product placement. We summarize the PE-score algorithm in Figure 1.

Two remarks on the algorithm are in order. First, we train a 3D CNN on video content data using video-level engagement as the outcome variable. The number of parameters to estimate far exceeds the

Figure 1. Summary of the PE-Score Algorithm

training sample size. This is a common feature of deep learning models and does not necessarily imply overfitting (e.g., Zhang et al. 2017). Nevertheless, we take several actions to mitigate overfitting concerns. We (1) train the algorithm using a large sample of videos; (2) use transfer learning to reduce the number of parameters to estimate; (3) use effective regularization methods, such as dropout (Srivastava et al. 2014) and early stopping; and (4) check for overfitting on the validation sample. Reassuringly, as we discuss in Online Appendix A, the test results suggest no signs of overfitting.

Second, we want to predict sales lift from the content of influencer video ads. We do so using the PE score as an interim summary statistic that is succinct and interpretable. This helps us understand what lifts sales, which in turn, helps practitioners approach the common problem of marketing attribution (e.g., Testwuide 2020). Moreover, the PE score is constructed without sales data and is evaluated on its ability to predict sales lift out of sample. This is arguably a more stringent test than testing the predictive power of a model that uses sales as the outcome variable both in and out of sample.

In what follows, we evaluate the PE-score algorithm. Specifically, we test whether influencer video ads with higher PE scores lift sales more. We present the data next.

4. Data

We test whether the PE-score algorithm predicts sales lift. To do so, we need data on influencer video ads, video engagement metrics, and sales revenue of the advertised products. We developed such a data set via collaboration with an entertainment commerce company. For context, at the time of this study,

content and engagement data were usually stored in one system (i.e., social media platforms, such as TikTok), whereas sales data were typically stored in another (e.g., e-commerce websites, such as Taobao). It is valuable to be able to connect these data sources.²⁷

We collect influencer video ads data from the Chinese version of TikTok because of its mature ecosystem around influencer video advertising. There is an established marketplace called Xingtu, where sellers contract with influencers to advertise their products. To date, Xingtu has attracted over 2 million influencers and 1.9 million registered sellers.²⁸ Two notable features characterize this marketplace. First, engagement is the centerpiece of the ecosystem. It determines how influencers price their video ads, how sellers search for influencers, and how sellers monitor ad performance. Second, influencers have significant discretion in designing video ad content. In a typical ad creation process, a seller provides some general guidelines, and an influencer drafts an ad script, makes the video upon seller confirmation of the script, and posts the ad upon seller confirmation of the video. Sellers are able to influence ad content to some degree. However, there are many video design aspects that are controlled by the influencer. In particular, there is no clear way for sellers to predict sales lift from an ad. They pay for engagement in the (sometimes unfulfilled) hope that engaging influencers will lift sales.

To further understand the TikTok influencer advertising market, we interviewed a number of practitioners in this space. Online Appendix D presents the scripts. These interviews suggest that, indeed, (1) sellers do not tend to influence the visual aspect of video content that we focus on in the paper; (2) sellers do not tend to influence product placement in the video ad; and (3) influencers do not tend to choose the

posting time of video ads based on product-specific demand, a fact that we will later use for identification.

We capture sales data on Taobao. Taobao is the biggest e-commerce website in China.²⁹ The vast majority of sellers in our video ads data list their products on Taobao exclusively as indicated by the product link in the video ads. We also confirmed with the partner company that TikTok and Taobao were indeed the main advertising and sales channels for sellers during the time of the sample. This helps us link product sales lift on Taobao to video ads on TikTok.

More specifically, the data set is a matched sample from two separate sources. The first is a video data set that contains all TikTok influencer video ads with product links from March to June 2019. For each influencer video ad, the data contain the video, its posting date, product identifier, engagement metrics, and influencer characteristics taken as a snapshot in January 2019.

The second source is a product data set of all products listed on Taobao from May to November 2019. For each of these products, the data track its product identifier, sales revenue on Taobao, product image, category, price, and discount. Following Taobao's policy, product revenue is reported as "30-day sales," meaning the sum of sales revenue over the previous 30 days, including the current day, measured at the daily level.³⁰ Some products are missing category information. We use product titles and nonmissing category labels to train a machine learning model that predicts the missing product categories. The model has an accuracy of 82% in the test sample (see Online Appendix E for details).

We match the two data sources using the product identifier.³¹ Among the products that have matching influencer video ads on TikTok, the majority of products have one video ad. We focus on these products in subsequent analysis to simplify attribution.³² This yields 2,734 influencer video ads with matching product sales data among influencer video ads from the first source. Among these 2,734 video ads, 2,685 have complete influencer characteristics. We call the panel data set of these 2,685 product-video pairs the *sales*

panel, which we will set aside to evaluate the PE-score algorithm's ability to predict sales lift.

The sales panel spans the window of May 1 through November 5, 2019. During this window, different products in the sales panel have different numbers of observed days. Some products might have been newly listed on Taobao during this window. Some products might have changed the product identifier. In addition, there are on average 16.8% missing sales observations for each product between its first observed days and last observed days in the sales panel, which the partner company believed to have happened for unrelated mechanical reasons. Figure F.1 in the Online Appendix shows the distribution of observed days in the sales panel. Most products have around 50–90 observed days.

Among the remaining videos from the first source, as discussed, we focus on videos with spoken words to control for the video transcript. This yields the aforementioned 16,951 video ads that we use to construct the algorithm: 10,000 for training the 3D CNN, 3,500 for validation, and the remaining 3,451 for a holdout test of the algorithm, all through random assignment. We call the data set of these 16,951 video ads the *construction sample*. Altogether, this paper draws on a total of $2,685 + 16,951 = 19,636$ video ads.

Note that we do not require the sales panel and the construction sample to be comparable. In fact, once the algorithm is constructed, PE scores of videos in the sales panel can be computed prior to release without relying on their engagement data. This feature contributes to algorithm scalability. As mentioned, it also offers a stringent test of the algorithm; we construct it from one sample of videos without any sales information (the construction sample) and test its predictive power on sales lift on a different sample (the sales panel), which helps examine the external validity of the algorithm.

Table 1 presents the summary statistics of observed video engagement metrics: the numbers of likes, comments, and shares.³³ Engagement takes time to grow. To capture each video's ultimate level of engagement,

Table 1. Summary Statistics of Observed Video Engagement

Variable	N	Mean	Standard deviation	Min	Median	Max
Videos in the sales panel						
Likes	2,685	38,515	111,116	0	3,654	1,831,709
Comments	2,685	542	2,052	0	84	71,068
Shares	2,685	936	5,007	0	80	166,821
Videos in the construction sample (i.e., training, validation, and test sets)						
Likes	16,951	34,339	112,302	0	3,021	2,553,627
Comments	16,951	531	2,124	0	69	71,068
Shares	16,951	1,184	6,690	0	91	195,563

Notes. The sample consists of all 19,636 videos used for this paper, which are divided into 2,685 videos in the sales panel and 16,951 videos in the construction sample, with no overlap between the two subsets. Each engagement metric is at the video level.

Table 2. Summary Statistics of Products and Influencers (Sales Panel)

Variable	N	Mean	Standard deviation	Min	Median	Max
Products						
<i>Average 30-Day Sales Revenue</i>	2,685	246,680	5,288,389	0	9,446	272,107,695
<i>Price</i>	2,685	1,081	39,220	0	68	2,019,515
<i>Discount</i>	2,685	100	506	0	20	13,901
Influencers						
<i>Gender</i>	1,404	0.58	0.49	0	1	1
<i># Followers</i>	1,404	1,617,806	3,048,990	0	723,679	43,012,100
<i>Average Play</i>	1,404	635,432	3,255,567	0	74,908	97,890,191
<i>Price per Video Ad</i>	1,404	19,530	53,808	0	6,000	1,000,000
<i>Expected Cost per Mille (CPM)</i>	1,404	1,027	21,315	0	121	785,714
<i># Video Ads Influencer Has Posted</i>	1,404	13	27	0	2	265

Notes. The sample consists of all products in the sales panel and their associated influencers. Each product-related variable is at the product level and is measured in RMB. A product's average 30-day sales revenue is taken over its observed days in the sales panel. Price and discount contain no variation at the product level over the duration of the sales panel. Each influencer-related variable is at the influencer level and was recorded in January 2019. Gender is an indicator variable that equals zero for female influencers. Price per video ad is in RMB. CPM is the cost per mille (1,000) plays in RMB.

we use its last observed value in our data, which occurs, on average, 28 days after posting. These engagement metrics are statistically indistinguishable between the sales panel and the construction sample except that videos in the sales panel were shared less on average.

The upper portion of Table 2 presents the summary statistics of sales revenue, as well as prices and discounts, of all products in the sales panel. Their average 30-day sales revenue is 246,680 Chinese yuan (RMB) or 35,699 U.S. dollars based on the average 2019 currency exchange rate of 6.91:1. The lower portion of Table 2 presents the summary statistics of all influencers in the sales panel. Each influencer posted 1.9 video ads in the sales panel on average.

Online Appendix F presents further details of the sales panel. In summary, engagement and sales show sizable variation across videos (Figure F.2 in the Online Appendix). Most influencers post one video ad, although there is a distribution (Figure F.3 in the Online Appendix). The most common video length is 15 seconds, whereas the video posting date spans the window of the sales panel (Figure F.4 in the Online Appendix). The most common category in the data is food followed by makeup (Figure F.5 in the Online Appendix).

A pattern to note in the data is the lack of a significant correlation between video engagement and

changes in sales. Figure F.6 in the Online Appendix presents the scatterplots of the relationship between the observed engagement metrics (the numbers of likes, comments, and shares, respectively) and the difference in average 30-day sales before and after a video ad is posted. Sales difference has no significant correlation with observed engagement ($\rho = -0.0075$, $p = 0.90$ for likes; $\rho = -0.02$, $p = 0.74$ for comments; and $\rho = -0.0074$, $p = 0.91$ for shares). This result suggests that using video engagement to evaluate ad effectiveness can be misleading. The PE-score algorithm is intended to address this problem. We present its evaluation in the next section.

5. Algorithm Evaluation

In this section, we first present the computational results of the PE-score algorithm. We also show suggestive evidence of the argument that influencers and sellers may have misaligned incentives. We then proceed to the main test of whether influencer video ads with higher PE scores lift sales more effectively.

5.1. Computational Results of the Algorithm

For each video ad, the algorithm outputs a 3D engagement heat map, a 3D product heat map, and a PE score. Table 3 presents video-level summary statistics

Table 3. Summary Statistics of the Video-Level Computed Engagement Score, Product Score, and PE Score (Sales Panel)

Variable	N	Mean	Standard deviation	Min	Median	Max
<i>Engagement Score</i>	2,685	0.48	0.15	0.00	0.49	1.00
<i>Product Score</i>	2,685	0.18	0.14	0.00	0.15	1.00
<i>PE Score</i>	2,685	0.21	0.14	0.00	0.19	1.00

Notes. The sample consists of all videos in the sales panel. All three scores in the table are at the video level and normalized to $[0, 1]$ over all videos in the sales panel. Each video's engagement score and product score are computed as the sum (rather than the average) of their respective pixel-level values in the video to reduce correlation with the PE score for subsequent analysis.

of these three outputs for videos in the sales panel, which is the sample that we will use to evaluate the algorithm. In Table 3, the “engagement score,” termed this way to differentiate it from observed engagement, is a video’s sum of pixel-level engagement values. The “product score” is a video’s sum of pixels in which the product appears.³⁴ To facilitate interpretation, we normalize all three scores to the interval of $[0, 1]$ in Table 3 and in subsequent analyses.

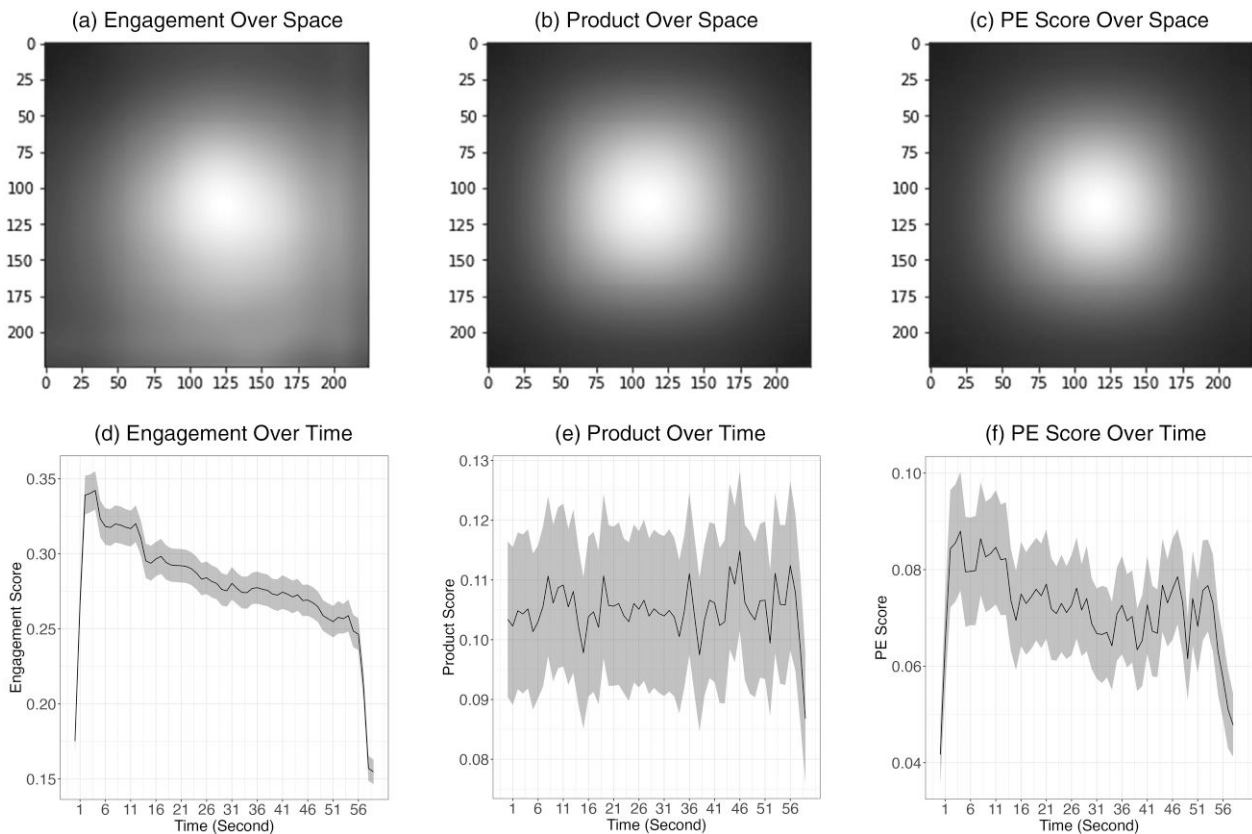
To further visualize these computational results, we present average pixel-level engagement and product scores within a video frame (Figure 2, (a) and (b)) and over the duration of a video (Figure 2, (d) and (e)). For completeness, we analogously present average pixel-level PE scores, computed as the pixel-level multiplication of engagement and product scores, over either space (Figure 2(c)) or time (Figure 2(f)).

On average, the most engaging region of a video frame tends to be its center, with the bottom and right sides of the screen being slightly more engaging than the top and left sides. This is possibly because the bottom is where the information of the video (e.g., the influencer name) is shown and because the right is

where engagement metrics are shown. Similarly, on average, products tend to appear in the center of the frame, and pixel-level PE scores also tend to peak around the center. However, we cannot simply conclude that we should put the product in the center. The most engaging regions of a video vary from frame to frame. The average structural similarity index measure (SSIM) between two consecutive engagement heat maps in the data is 0.78.³⁵ More importantly, products do not always appear in high-engagement pixels. The SSIM between the engagement heat map and the product heat map on the same frame averaged over all videos in the sales panel is 0.46, which is moderate.

Over the duration of a video, engagement tends to start low, rise rapidly, peak in the first 6 seconds, decline gradually from the 7th to 57th seconds, and decline very sharply in the last 3 seconds. Product placement is noticeably different; it tends to be uniform, except in the last 3 seconds. The PE score follows a pattern similar to engagement; it rises, then falls, and falls sharply near the end of the video, possibly because of fading engagement in these moments.

Figure 2. Distribution of the Computed Engagement Score, Product Score, and PE Score



Notes. The sample consists of all videos in the sales panel. All three scores in the figure are at the video-second level and normalized to $[0, 1]$ over all video-seconds in the sales panel. In images (a)–(c), brighter means higher engagement, more product placement, and higher PE scores, respectively. In images (d)–(f), the gray areas represent values within 0.1 standard deviations from the mean value in that second.

However, we again cannot simply conclude that products should be placed in moments when average engagement peaks. These dynamics vary significantly across videos. The gray areas in Figure 2, (d)–(f) represent values within 0.1 standard deviations from the mean, which span a noticeable range already. These observations highlight the incremental value of the PE-score algorithm, which captures rich heterogeneity across space, time, and videos. In the next section, we use the computed PE score to present suggestive evidence of its underlying argument— incentive misalignment.

5.2. Incentive Misalignment

At the time of this study, influencers were typically paid a fixed price per video ad, which was mostly driven by the number of followers and engagement (Li et al. 2024 also noted that influencers' primary goal is engaging their audience). Moreover, product ads during entertainment are generally disliked (e.g., Elpers et al. 2003, Wilbur 2016); influencers may even lose followers by posting sponsored videos (Cheng and Zhang 2024). Therefore, influencers may have more incentive to optimize a video ad for engagement rather than sales. In light of the PE-score concept, this means that influencers may not want to place the product in the most engaging spots of the video. Figure 3 presents an example from the data. The algorithm reveals that the most engaging spots of the video-ad frame feature the influencer's face, not the advertised product.

It is important to note that even though sellers can fully observe a video ad, they may have different interpretations of the content than the influencer. For example, sellers may not know what engages a particular influencer's followers. This is similar to how medical notes or legal documents may be fully observable to

both doctors and patients or lawyers and clients, but information asymmetry can still arise because of the asymmetry in knowledge. This information asymmetry is also a reason why sellers ask influencers to design video ads in the first place.

We supplement this discussion with a smell test of the incentive misalignment argument. We collected a separate sample of 77 video ads where influencers advertised their own products, and we compare them with the 2,685 videos in the sales panel. If the incentive argument is true, these 77 video ads should have higher PE scores than those in the sales panel.³⁶ We pool these two types of videos and regress the PE score on an indicator variable of whether the influencers are advertising their own products while controlling for the product and influencer characteristics in Table 2 as well as the product-category indicators.³⁷ We find that PE scores are, on average, 35% higher ($p < 0.001$) when influencers are advertising their own products.

Finally, our argument is that influencers are *able* to design effective video ads but may act differently for strategic reasons. To test this argument, we examine the effect of influencers' experience. If the lack of ability, as opposed to incentive, is what hinders advertising effectiveness, influencers with less experience should produce lower PE scores. We regress the PE score on measures of influencer experience, including the number of video ads that the influencer has posted and the number of days since the influencer's first post, controlling for other influencer characteristics in Table 2.³⁸ We find no statistically significant association between the PE score and influencer experience measures.

Taken together, the evidence is consistent with the incentive misalignment argument. The PE score can help mitigate this problem by quantifying to what

Figure 3. An Example of Incentive Misalignment



Notes. The left image shows a frame from an influencer video ad in the data. The center image shows its corresponding frame in the engagement heat map, where brighter areas are more engaging (see Online Appendix B). The right image shows its corresponding frame in the product heat map, where bright areas indicate product presence (see Online Appendix C).

extent the influencer is effectively advertising the product. We test the PE-score algorithm next.

5.3. Influencer Video Ads with Higher PE Scores Lift More Sales

We begin by presenting model-free evidence that video ads of higher PE scores are more effective. We then evaluate the PE-score algorithm in two steps. First, we identify to what extent each video ad lifts sales. Second, we quantify to what extent a video's PE score predicts its sales lift, among other predictors. This two-step approach is analogous to the standard DID with heterogeneous treatment effects, which combines identification and prediction in one step in a linear model. By separating identification and prediction, the two-step approach allows each step to be more flexibly implemented. For example, one can use Extreme Gradient Boosting (XGBoost) or any machine learning model to explore predictors of sales lift.³⁹

5.3.1. Model-Free Evidence. We first present the model-free relationship between the PE score and the difference in average 30-day sales revenue before and after a video ad is posted (i.e., treated). Among the 2,685 products in the sales panel, 259 were treated during the data window. We calculate each of these 259 products' sales difference as its average 30-day sales revenue within the data window after the posting of its video ad minus its average 30-day sales revenue before. We plot the sales difference against the PE score of the corresponding video. Figure 4 shows the scatterplot. There is a positive and significant correlation between

the PE score and sales difference ($\rho = 0.16$, $p < 0.01$), consistent with our hypothesis. In contrast, as discussed, engagement has no significant correlation with sales difference (Figure F.6 in the Online Appendix).

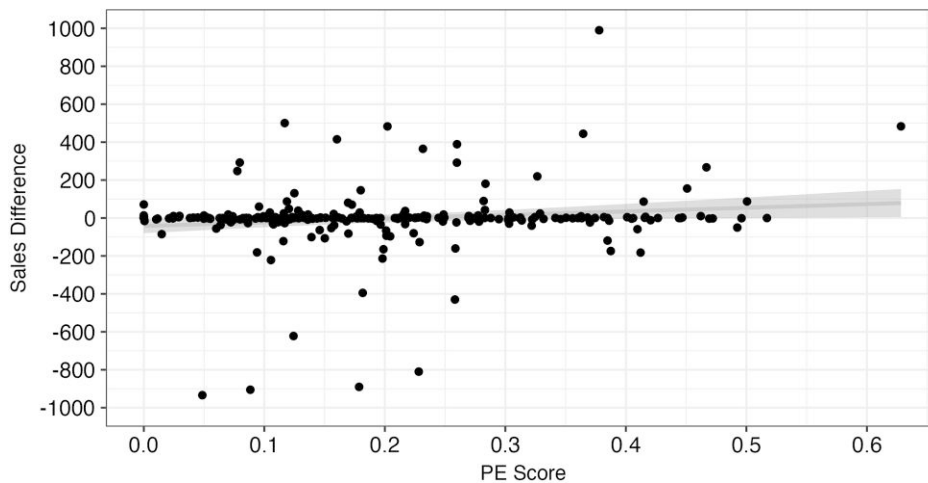
5.3.2. Identifying Video-Specific Sales Lift. As the first step in our formal evaluation of the algorithm, we identify the video-level sales lift from posting an influencer video ad. We use DID for identification, drawing on the fact that different products in the sales panel posted video ads at different times (e.g., Stevenson and Wolfers 2006, Liu et al. 2019). As mentioned, among the 2,685 products in the sales panel, 259 posted video ads within the observed time window; we call them "treated products." The remaining 2,426 products posted videos before the observed time window and thus, experienced no treatment event during the time window; we call them "control products." The DID method relies on the assumption that treated and control products have comparable time trends absent the treatment event. We discuss this assumption and show the robustness of the results with respect to alternative control-group definitions in Online Appendix H.2.

We identify each treated product's sales lift by estimating the following clustered ordinary least squares (OLS) specification:

$$\text{Daily Sales}_{vd} = \alpha \cdot \text{Post}_{vd} + \text{Video}_v + \text{Day}_d + \gamma \cdot \text{Search}_{vd} + \epsilon_{vd}. \quad (2)$$

As discussed, we focus on products that have only one video ad so that the subscript v indexes both the

Figure 4. Before-After Sales Difference by PE Score



Notes. The sample consists of the 259 products in the sales panel that had a video ad posted during the data window. The x axis is a product's computed PE score. The y axis is a product's sales difference, which equals the product's average 30-day sales revenue (in 1,000 RMB) after posting its video ad minus its average 30-day sales revenue before. We restrict the y axis to values between −1,000 and 1,000 for visualization, and the resulting figure displays over 97% of the treated products. The gray areas represent the 95% confidence band along the regression line. Over all of these 259 products, the correlation between the PE score and the sales difference is 0.16 ($p < 0.01$); the linear regression of sales difference on the PE score and an intercept yields a PE-score coefficient of 1,243 ($p < 0.01$).

video and the product. The subscript d indexes the calendar day.

The dependent variable $Daily\ Sales_{vd}$ is the imputed daily sales revenue of product v on day d . As explained, we only observe each product's 30-day sales revenue. For cleaner attribution, we impute each product's daily sales revenue from its 30-day counterpart.⁴⁰ Drop the product subscript v for now, and let $t = 1$ denote the first day that this product is observed in the sales panel. For the 30-day “initialization period” of $t \in \{-28, -27, \dots, -1, 0, 1\}$, we assume $Daily\ Sales_t = 30\text{-DaySales}_1 / 30$. We then compute daily sales for each $t \geq 2$ recursively as $Daily\ Sales_t = 30\text{-DaySales}_t - 30\text{-DaySales}_{t-1} + Daily\ Sales_{t-30}$.⁴¹ This imputation rule is based on the partner company's observation that sales tend to be stable in this market unless there are promotional events. The results are robust if we instead allocate 30-DaySales_1 in proportion to product search volume (more details to follow) over the initialization period, if we exclude products that experienced unusual fluctuations in search volume during the initialization period, or if we use the Denton–Cholette method to disaggregate each product's 30-day sales time series into daily sales (Online Appendix H.3).

The treatment variable is $Post_{vd}$, which equals one if video ad v is posted by day d and equals zero otherwise. Among the 2,685 products in the sales panel, the 259 treated products experienced a change of $Post_{vd}$ from zero to one during the observation window. The remaining 2,426 control products posted videos before the observation window so their $Post_{vd}$ is always one (coding $Post_{vd}$ as always zero or one does not affect the estimation results). This specification estimates the average daily sales lift since the posting of a video ad. The results are robust if $Post_{vd}$ is defined to be one on day d when video ad v is posted and zero otherwise or if we consider the truncated nature of sales observations after ad posting (Online Appendix H.2).

Using the panel structure of the data, we include video/product fixed effects $Video_v$ to control for unobserved heterogeneity across videos/products. We also include day fixed effects Day_d to capture common time effects (e.g., trends, seasonality) on sales. We can in theory include influencer fixed effects, but their magnitude will not be separately identified from video fixed effects because each video in our data has only one associated influencer.

Reverse causality could be a concern if an influencer posts a video ad in anticipation of sales lift (for instance, if the product is being advertised on other channels). This concern may not apply in the paper's setting because as discussed, influencers are mainly motivated by engagement metrics instead of product sales. Moreover, according to the partner company, many sellers at the time of this study are small sellers who have limited advertising resources. Our practitioner interviews also

suggest that influencers do not tend to choose ad posting time based on product-specific demand (Online Appendix D). Nevertheless, we collect the Baidu Search Index of each product in the sales panel as a proxy for its unobserved time-varying demand shifters and include it as a control variable, $Search_{vd}$ (Online Appendix G).⁴² The parameter γ captures the effect of search. The error term ϵ_{vd} is clustered by video/product and day.

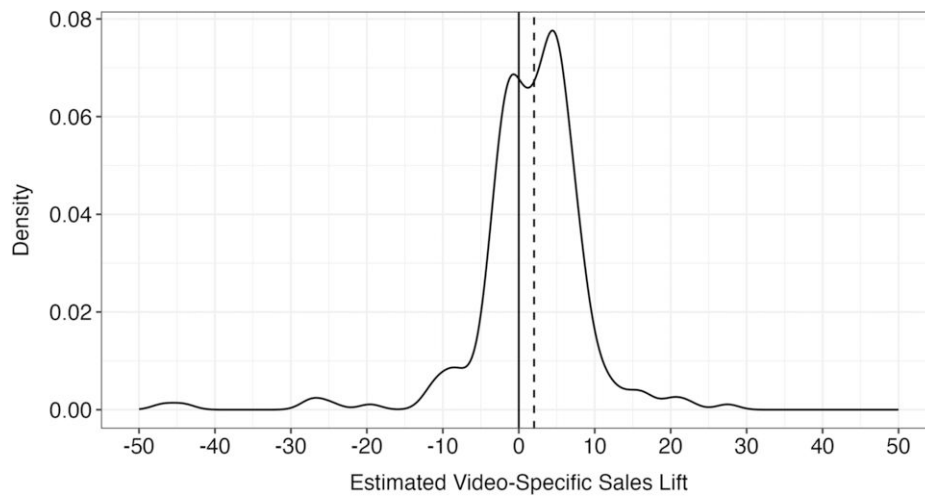
The parameter of interest in this DID estimation is α , which measures the treatment effect or sales lift from a video ad. We separately estimate the sales lift of each of the 259 treated products using the remaining 2,426 products as controls. For each treated product, we extract its corresponding record from the sales panel and combine it with the records of the 2,426 control products in the sales panel. We estimate the sales lift of this treated product from this combined panel data of $1 + 2,426 = 2,427$ products. We repeat the DID estimation 259 times to obtain 259 sets of parameter estimates.

Figure 5 shows the distribution of the 259 sales lift estimates across videos. Sales lift is concentrated around zero, with more mass on the positive side and long tails on both sides, qualitatively consistent with previous findings on the distribution of ad treatment effects on social media (e.g., Wernerfelt et al. 2024).

Table 4 presents the summary statistics of the coefficients from the DID estimation. The average sales lift (α) is 2,017 RMB or \$292, not significantly different from zero ($p = 0.50$). This result is worth noting given that sellers pay nontrivial amounts to advertise their products; influencers in our data charge 19,530 RMB or \$2,826 per video ad on average (Table 2). It will be helpful to be able to predict sales lift before investing in an influencer video ad, an issue that we will examine in the next section. In addition, search intensity shows a positive association with sales (γ) across all treated products, and its mean association of 0.41 is significantly different from zero ($p < 0.001$). To address the possibility that search fully mediates the ad effect leading to a spuriously insignificant average sales lift, we repeat the DID estimation without search as a control variable. The average sales lift remains insignificantly different from zero ($p = 0.56$).

5.3.3. PE Score Predicts Sales Lift. In the second step of algorithm evaluation, we test the predictive power of the PE score on sales lift, among other predictors. We begin with OLS specifications where the dependent variable is the video-specific sales lift estimated from step 1 and the independent variables are various sets of predictors. Results are robust if we use bootstrapping to account for the measurement uncertainty in sales lift (see Online Appendix H.4).

Figure 5. Distribution of Sales Lift Across Videos



Notes. The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. Sales lift is estimated at the product level, measured in 1,000 RMB. The vertical solid line marks zero. The vertical dashed line marks the average sales lift at 2.02. For visualization, the range of the x axis is restricted from -50 to 50 , about one standard deviation of the sales lift estimates (see Table 4).

As column (1) in Table 5 shows, the PE score is a positive and significant predictor of sales lift ($p < 0.01$). The PE score alone explains 3% of the variance in sales lift. This result is consistent with our hypothesis that influencer video ads with higher PE scores are more effective at lifting sales.

To test whether it is product-related engagement that predicts sales lift, we check whether overall engagement *alone* or product placement *alone* would have predictive power. As column (2) in Table 5 shows, the engagement score, computed as each video's sum of pixel-level engagement values as explained in Section 5.1, is an insignificant predictor. If we replace the computed engagement score with observed engagement (the number of likes, comments, or shares), its coefficient is also insignificant (Online Appendix H.5). This result reaffirms industry observations and the correlational finding in Section 4 that engagement does not necessarily predict sales lift. Column (3) in Table 5 further shows that the product score, computed as each video's sum of pixel-level product placement as explained in Section 5.1, is also an insignificant predictor. In other words, simply showing the product more in the video may not predict better sales.

We also examine video ads that both engage the viewer and feature the product actively, with the two

activities potentially conducted in isolation. For example, a video may be entertaining in the first half and feature the product intensively in the second half, resulting in high engagement and high product scores but a low PE score. We examine the interaction term between the engagement score and the product score as an alternative to the PE score. This term is also insignificant, as shown in column (4) of Table 5.

Column (5) in Table 5 shows the estimation result when the PE score, the engagement score, the product score, and the interaction between the latter two scores are simultaneously included. The PE score remains the only significant predictor ($p < 0.001$), and its effect remains positive. These findings together suggest that improving engagement and product placement separately may not help; it is important that these two are aligned spatiotemporally as captured by the PE score. This result echoes Zhang et al. (2020), who found that the temporal synchronicity between user-comment volume and movie content predicts movie enjoyment.

Last, we build on the specification in column (5) of Table 5 and add a rich set of covariates, namely influencer and product features as reported in Table 2, the product's average Baidu Search Index over the observation window, and the product-category indicators. Column (6) of Table 5 shows the results. The PE score

Table 4. Summary Statistics of DID Estimation Results

Parameter	N	Mean	Standard deviation	Min	Median	Max
Sales Lift (α)	259	2.02	47.53	-123.62	2.06	699.20
Search Coefficient (γ)	259	0.41	0.01	0.39	0.41	0.49

Notes. The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. Sales lift is estimated at the product level and is in 1,000 RMB.

Table 5. Predicting Sales Lift

	Dependent variable: <i>Sales Lift</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
Computed scores						
<i>PE Score</i>	68.27** (22.87)				101.32*** (27.04)	100.79*** (28.91)
<i>Engagement Score</i>		13.79 (20.20)			2.21 (30.55)	−2.44 (32.36)
<i>Product Score</i>			−6.05 (21.82)		−106.22 (76.98)	−112.47 (80.46)
<i>Engagement Score × Product Score</i>				7.10 (40.25)	98.61 (149.97)	126.50 (156.90)
Influencer features						
<i>Gender</i>						5.86 (6.37)
<i># Followers (in Millions)</i>						0.24 (1.49)
<i>Average Play (in Millions)</i>						0.31 (2.32)
<i>Price per Video Ad (in Millions)</i>						−9.04 (183.65)
<i>Expected CPM (in Millions)</i>						13.04 (43.58)
<i># Video Ads Influencer Has Posted</i>						−0.01 (0.18)
Product features						
<i>Average Search</i>						−3.67 (7.10)
<i>Price</i>						0.003 (0.01)
<i>Discount</i>						−0.002 (0.005)
Product-category indicators	No	No	No	No	No	Yes
Observations	259	259	259	259	259	259
R^2	0.03	0.002	0.0003	0.0001	0.06	0.08
Adjusted R^2	0.03	−0.002	−0.004	−0.004	0.04	−0.005

Notes. The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level or video-level sales lift in 1,000 RMB.

** $p < 0.01$; *** $p < 0.001$.

continues to be a positive and significant ($p < 0.001$) predictor of sales lift.⁴³ In fact, it is the only significant one among all predictors, including the unreported product-category indicators. The inclusion of the covariates improves R^2 but hurts adjusted R^2 (which considers the number of predictors). This result suggests that the PE score is a better predictor of sales lift than influencer characteristics, such as the number of followers, which are commonly used to select influencers for advertising. We show that popular influencers and expensive influencers may not be more effective. What they do in the video ad may be more important than who they are.

One fact to notice about Table 5 is the relatively low R^2 measures in all columns, possibly because of the

functional restrictions of OLS. To evaluate the predictive power of the PE score under more flexible specifications, we use XGBoost (Chen and Guestrin 2016),⁴⁴ a popular decision-tree based predictive-modeling algorithm, to predict sales lift with the same variables as in column (6) in Table 5. We train the XGBoost model on the 259 treated products. We set the number of boosting rounds to a small value of $n = 2$ to mitigate overfitting concerns. The XGBoost model indeed fits better than the OLS model, achieving an R^2 of 0.47 and an adjusted R^2 of 0.42.⁴⁵

Table 6 shows the most important features in the XGBoost model ranked by “gain.” Gain is the standard metric to evaluate feature importance in decision trees (e.g., Zheng et al. 2017). It refers to the average

Table 6. XGBoost Feature Importance in Predicting Sales Lift

Feature	Gain	Cover	Frequency
PE Score	0.865	0.338	0.242
Expected CPM	0.044	0.075	0.091
Product Score	0.026	0.329	0.152
Engagement Score	0.025	0.006	0.091
Discount	0.017	0.034	0.212
# Followers	0.013	0.070	0.091
Average Search	0.009	0.143	0.061
Price	0.0002	0.002	0.030
# Video Ads Influencer Has Posted	0.0001	0.003	0.030

Notes. The sample consists of all treated products in the sales panel. The features are ranked by gain, the standard metric of feature importance for the XGBoost model’s predictive accuracy. The table reports up to 10 of the most important features of the XGBoost model trained on the entire sample. When the number of features shown in the table is less than 10, the unshown features are not used by the XGBoost model to split nodes, meaning that they are unimportant according to the model.

improvement in model accuracy after a node is split based on a feature. In the context of this paper, gain quantifies how much a feature contributes to the XGBoost model’s accuracy in predicting sales lift, and it ranks the PE score as the most important predictor by far. We also report the “cover” and “frequency” metrics for completeness. Cover is the fraction of the sample for which a feature is used to split a node. Frequency is the fraction of times that a feature is used to split a node. Cover and frequency indicate how often a feature is used, without necessarily reflecting the feature’s contribution to the model’s predictive accuracy. Nevertheless, the PE score has the highest cover and frequency among the features in the XGBoost model.

We conduct extensive robustness analyses. Online Appendix H presents the details. In summary, the PE score’s ability to predict sales lift is robust with respect to alternative ways to construct the engagement heat map (Online Appendix H.1), to causally identify sales lift (Online Appendix H.2), to impute daily sales (Online Appendix H.3), to handle measurement uncertainty in sales lift (Online Appendix H.4), to define engagement as a predictor of sales lift (Online Appendix H.5), and to compute the engagement score and the product score as predictors of sales lift (Online Appendix H.6).

6. Exploring Drivers of the PE Score

We have shown that the PE score varies across videos, and this variation matters in predicting sales lift. We supplement the main analysis by exploring what substantive measures an influencer can take to improve the PE score. Recall that the PE score captures the spatio-temporal synchronicity between the engagement and

product heat maps. The product heat map has a straightforward interpretation; it captures product presence. The engagement heat map is less interpretable; it outputs the more-engaging regions of a video without offering a substantive explanation. We examine this issue next.

An established approach would be to use proven theories to guide the interpretation of unstructured data. For example, Zhang et al. (2020) used film grammar to analyze movie content, and Zhang et al. (2022) used photography theory to evaluate image quality. The challenge in our setting, as confirmed by the partner company, is that there is not yet a widely accepted theory on how to make influencer video ads engaging. Therefore, we explore drivers of engagement in a bottom-up, data-driven way. We do so at the pixel, frame, and video-segment levels in order.

Given the engagement heat map, the first question is what objects tend to appear in high-engagement pixels. To answer this question, we divide the pixels in a video into high and low types based on a median split of pixel-level engagement scores. Then, we create two versions of the video: one that only uses high-engagement pixels with low-engagement pixels blacked out (high version) and the reverse (low version). We then run an object detection algorithm, You Only Look Once or YOLO (Redmon et al. 2016),⁴⁶ on the high and low versions of the same video to identify what objects, from 80 prespecified classes, are presented in each version. For each detected object, we compute its net frequency of appearance in high versus low versions of all videos in the sales panel. Figure I.1 in Online Appendix I presents the results. More object instances are detected in high-engagement pixels (9,916) than in low ones (4,780). Moreover, as one would expect, humans are the most represented class in high-engagement pixels.

Based on the finding that human presence is a key part of engagement, the next question is what humans can do in the video to engage. Past research has identified the human face as an engaging object that attracts likes and comments on social media (e.g., Bakhshi et al. 2014, Li and Xie 2020, Hartmann et al. 2021). Indeed, as a sanity check of the algorithm, we find a positive and significant correlation between the presence of human faces and pixel-level engagement (Online Appendix B.2). We further ask what facial expressions drive engagement; facial expressions are arguably more actionable than factors such as facial attractiveness. We run an emotion detection algorithm, Facial Expression Recognition or FER (Zhang et al. 2016, Arriaga et al. 2017),⁴⁷ that detects facial expressions of the Ekman (1992) six basic emotions (anger, disgust, fear, happiness, sadness, and surprise) plus a neutral emotion. We apply FER to each frame of each video in the sales panel. Then, we regress the average pixel-level engagement scores on a given

frame on the probabilities of the detected emotions as output by FER. The results are shown in Figure I.2 of the Online Appendix. Sadness and happiness are positively and significantly associated with engagement scores, whereas fear and surprise are negatively and significantly associated. The pattern echoes earlier findings in the literature. For example, Wild et al. (2001) found that happiness and sadness are particularly contagious, taking effect in as short as half a second. The fast speed to engagement can be particularly helpful in short-form videos.

Last, to expand the space of actionable recommendations beyond facial expressions, we ask what actions an influencer can take in the video to engage. For a broad search of possible actions, we again take the data-driven approach by detecting what activities are in the video and how they relate to engagement. We divide each video in the sales panel into one or more segments that each last 15 seconds. We run an activity detection algorithm, Inflated 3D ConvNet or I3D (Carreira and Zisserman 2017),⁴⁸ on each segment to classify it into 400 prespecified classes.⁴⁹ Then, we regress the average pixel-level engagement score in a segment on the probabilities of the detected activities as output by I3D, controlling for segment sequence (e.g., the first in a video). We find 112 activities that are positively associated with engagement scores and 40 activities that are negatively associated with engagement scores. The top 30 activities by effect size that are significantly associated with engagement scores ($p < 0.05$) are reported in Figure I.3 of the Online Appendix. Positive activities tend to be more energetic and faster paced (e.g., side kick, salsa dancing, or krumping) or novel (e.g., disc golfing, getting a tattoo, or snorkeling), whereas negative activities tend to be slower paced (e.g., shearing sheep, weaving a basket, or filling eyebrows) or more mundane (e.g., garbage collecting, running on a treadmill, or reading books). These observations are consistent with Yang et al. (2022), who found a positive association between TV-ad tuning rate and ad energy related to arousal.

We examine these observations further in two ways. First, we look for underlying themes among these activities using topic modeling. Table I.1 in the Online Appendix lists the top 10 words in two topics for activities positively or negatively associated with engagement.⁵⁰ Words of higher energy, words of higher pace, and words that are possibly more novel (e.g., play, dance, or climb) are identified in more-engaging activities. Words of lower energy, words of lower pace, or words that are possibly more mundane (e.g., paper, question, or table) are identified in less-engaging activities.

Second, to overcome the relatively small sample that we have for the aforementioned topic-modeling exercise, we conduct an auxiliary survey with 101 college students and staff members at a university in

Beijing, China to identify the commonality in more-engaging activities versus less-engaging activities. These participants tend to be familiar with TikTok. As such, they may be able to interpret these activities in the context of TikTok beyond what topic modeling can reveal. Participants were asked to write three to five adjectives or phrases each to indicate their perceived common characteristics of the more-engaging activities versus less-engaging activities. We plot the word clouds of their responses in Figure I.4 of the Online Appendix. Participants tend to use words such as interesting, novel, and funny to describe more-engaging activities, whereas for less-engaging activities, they tend to agree on words such as boring, common, and physical labor.

Combining results from the analyses at the pixel, frame, and video-segment levels, we find that human presence, sad or happy emotions, and stimulating or novel activities are positively associated with engagement. To improve the PE score, it may be helpful to spatiotemporally align product placement with these elements of engagement. For example, it may be helpful to feature a product in a moment of high emotional connectivity or when an influencer is performing a stimulating activity. These recommendations echo the finding of Akpınar and Berger (2017) that it is possible and advisable to integrate the product into viral elements of an ad for better conversion; they cited Blendtec's "Will It Blend" campaign as a good example, where the blender famously shredded an iPhone. In our framework, shredding an iPhone is engaging, and the advertised product, the blender, is integral to the engagement, making it effective engagement.

7. Concluding Remarks

In this paper, we propose an algorithm to compute a metric called the PE score, which analyzes unstructured video data to predict the effect of influencer video ads on product sales. The PE score has an intuitive interpretation; it captures to what extent a product is shown in the most engaging part of the video or how engaging the video is when and where the product is shown.

We construct and evaluate the algorithm using a data set of TikTok influencer video ads and their corresponding product sales. The PE score is a significant predictor of the amount of sales revenue that a video ad is able to lift. The PE score much outperforms video engagement, product placement, influencer characteristics, and product features in predicting sales lift.

The findings are robust with respect to different ways to construct the algorithm, to measure sales lift, and to formulate its predictors. We also present suggestive evidence that incentive misalignment between influencers and sellers may explain the variation in the PE score. Last, we find that engagement increases with human presence, sad or happy emotions, and

stimulating or novel activities. It may be effective to integrate product placement with these engaging elements.

A practical advantage of the PE score is that it can be computed based on the already-trained PE-score algorithm before a video ad is released without relying on in-consumption data, such as eye movements or live interactions. As such, the algorithm can be used to evaluate a large number of candidate videos quickly, which helps with scalability, and to improve ad design prior to launch. The algorithm is also applicable beyond the placement of physical products in a video ad. The product can be replaced by a brand name, logo, or any key message that needs to be conveyed as long as the message is visually detectable in the video.

Various stakeholders in the influencer advertising space—and in the rising “creator economy” more broadly—can potentially benefit from the PE score. Influencers can use the PE score to aid video content development. They can adopt the drivers of engagement behind the PE score, place the product in the engaging pixels, or check a video ad’s PE score for feedback before release. In the era of generative artificial intelligence, the PE score can serve as a performance metric to automate video ad production. At the same time, sellers can use the PE score as a novel contractual instrument. For example, sellers can compensate influencers based on the PE score of their video ads. In comparison, the current industry norm of engagement-based compensation may exacerbate incentive misalignment, whereas sales-based compensation holds influencers accountable for product sales but exposes them to various factors beyond their control (such as perceived product quality, which is difficult to contract on). In this sense, the PE score can serve as a metric to help clarify the attribution of sales outcomes between sellers and influencers. Finally, entertainment-commerce platforms can use the PE score to launch various features to improve transaction efficiency. For example, a platform can highlight the PE score as a key performance index of influencers. Providing the PE score alongside engagement metrics can help sellers choose influencers and manage campaigns with richer information.

There are several directions for future research. First, it will be interesting to study various applications of the algorithm and track their impact on entertainment commerce. Second, the PE score is learned mainly through the visual components of a video ad while controlling for the acoustic features and spoken content. Although a similar PE score can be computed solely based on the sound, how to integrate the two in an interpretable way is a worthy topic. Data permitting, it is also worthwhile to examine an end-to-end model of the effect of ad content on sales. Third, our

exploration of engagement drivers is preliminary. Further studies including controlled experimentation may enrich the insight. Last, it will be meaningful to explore the generalizability of the PE score. We validate the algorithm in the context of influencer video ads, where the PE score likely matters because it captures the importance of attention in entertainment commerce and because influencers may want to draw attention to themselves. However, the general principle of product-oriented, effective engagement should extend to other forms of advertising. It will be useful to test whether the algorithm predicts sales lift based on mere ad content, being that a TV ad or a live-stream marketing session (e.g., Liu et al. 2022).

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Endnotes

¹ See <https://twitter.com/conniechan/status/1266476997699493889>.

² This integration of entertainment and commerce is happening in both directions. On one hand, social media platforms, such as Facebook, Instagram, and YouTube, are introducing shoppable content. On the other, e-commerce platforms, such as Alibaba, Amazon, and Walmart, are adding entertainment features.

³ See “TikTok revenue and usage statistics (2024)” (*Business of Apps*, October 10, 2024), <https://www.businessofapps.com/data/tik-tok-statistics>.

⁴ See “The state of influencer marketing 2024: Benchmark report” (*Influencer Marketing Hub*, September 16, 2024), <https://influencermarketinghub.com/influencer-marketing-benchmark-report>.

⁵ The company that provided us the data emphasizes strong industry demand for such predictive tools.

⁶ See “One million likes but less than 5,000 monthly sales, what did the product do wrong in short video marketing?” (*CAAS Data*, May 17, 2020), <https://www.niaogebiji.com/article-36539-1.html>.

⁷ See “B2B influencer marketing research shows a disconnect between brands and influencers” (*OST*, January 2, 2019), <https://ostmarketing.com/b2b-influencer-marketing-research-shows-disconnect-between-brands-and-influencers>.

⁸ Owned by Alibaba, Taobao is one of the world’s largest e-commerce websites and the major platform on which products advertised in TikTok influencer videos were sold during the time of our data.

⁹ The product image can be sourced from an e-commerce website where the product is being sold, but it can also simply be a cropped image from the video when the product is shown.

¹⁰ See “The state of influencer marketing 2024: Benchmark report” (*Influencer Marketing Hub*, September 16, 2024), <https://influencermarketinghub.com/influencer-marketing-benchmark-report>.

¹¹ Another closely related, growing line of research uses image data to inform various aspects of marketing, including social media engagement (Li and Xie 2020), brand image extraction (Liu et al. 2020), brand selfies (Hartmann et al. 2021), logo design (Dew et al. 2022), listing image design (Zhang et al. 2022), product aesthetics (Burnap et al. 2023), product-returns management (Dzyabura et al. 2023), facial image mining (Tkachenko and Jedidi 2023), labor-market research (Troncoso and Luo 2023), and business-survival prediction (Zhang and Luo 2023).

¹² The emphasis on scalable video analysis echoes Li et al. (2019), one of the first video-mining papers in marketing. Their paper highlighted the use of visual variation and video content, measures that can be automatically extracted from videos to explain crowd-funding outcomes.

¹³ See “TikTok ads: Everything you need to know about marketing on TikTok” (*Oberlo*, November 21, 2019), <https://web.archive.org/web/20200615002758/https://www.oberlo.com/blog/tiktok-ads>.

¹⁴ The seminal paper of Mitchell and Olson (1981) found that consumers’ attitude toward an ad can mediate their attitude toward the advertised brand. Our hypothesis complements their theory; we argue that attitude toward the ad, as measured by engagement, has a greater influence on attitude toward the brand if the brand is advertised in a more-engaging way.

¹⁵ ImageNet (<http://www.image-net.org>) is a database of over 1 million images with 1,000 class labels. It is considered the industry standard for training and testing image classification algorithms. Xception is an effective network for image classification, with a top-one accuracy of 0.79 and a top-five accuracy of 0.95.

¹⁶ See “Shares will matter more than likes, comments, or followers in the future of social media” (*Medium*, March 22, 2024), <https://medium.com/@commerciallybinu21/shares-will-matter-more->

[than-likes-comments-or-followers-in-the-future-of-social-media-5513a84d2776](https://medium.com/@commerciallybinu21/shares-will-matter-more-than-likes-comments-or-followers-in-the-future-of-social-media-5513a84d2776).

¹⁷ The nonvisual features can be used as side features and combined with visual features for joint training. This would account for the interaction between visual and nonvisual features. We follow the residual approach to train the 3D CNN only on visual features for the ease of extracting saliency maps. See Fong et al. (2024) for an in-depth study of music and Rajaram and Manchanda (2024) for a joint analysis of multiple video modalities.

¹⁸ For example, it is important to note that the PE score measures the engagement of product placement or presentation rather than how engaging the product itself is by design. Product design is often fixed by the time that influencers are making video ads. Product fixed effects help control for the possibility that some products are more engaging by design.

¹⁹ This is the standard image size for many widely used computer vision algorithms. However, the PE-score algorithm should accommodate any image size in principle.

²⁰ There is a paper famously titled “I just ran four million regressions” (Sala-i Martin 1997). Here, we just ran one regression with over 7 million parameters.

²¹ See <https://keras.io>.

²² For color images, the derivatives are with respect to increasing the value of a pixel along a particular color channel, which means increasing its intensity or brightness. For an interactive example, see <http://www.cknuckles.com/rngsliders.html>.

²³ See <https://github.com/keisen/tf-keras-vis>.

²⁴ Similar to standard regressions, this approach requires sufficient spatiotemporal variation in pixel values, which is likely satisfied given the high variety of videos on TikTok.

²⁵ See https://docs.opencv.org/4.x/da/df5/tutorial_py_sift_intro.html.

²⁶ Usually, a ratio test is performed on each matched key point to access its quality. The idea is the following. For a given key point, multiple matches with different distances can be found. One way to determine if the best match (the one with the shortest distance) is a good match is by looking at how it compares with the second-best match. If the two are too similar, the best match is more likely to be noise. If the two are different enough, the best match is more likely to be distinctive and thus, a good match. Following convention, we use a cutoff value of 0.75 and consider the product to be present at a given pixel if the ratio between the best match and the second-best match is below this cutoff.

²⁷ For instance, Lee et al. (2018) studied advertising content and engagement on Facebook, and they highlighted the lack of access to sales data as a limitation.

²⁸ See “China influencer marketing with Douyin and Xingtu” (*Ocean Engine*, May 31, 2023), <https://www.oceanengine.io/resource/china-influencer-marketing-douyin-xingtu-tips-strategies>.

²⁹ See “Top 15 Chinese eCommerce websites (updated 2024)” (*TMO Group*, February 8, 2024), <https://www.tmogroup.asia/insights/top-15-chinese-e-commerce-websites>.

³⁰ See a discussion of this policy at <https://zhidao.baidu.com/question/2202046371457180068.html>.

³¹ Some products might have changed their identifiers during the data window, which prevented us from matching every video with the corresponding product. More generally, the sales panel may not be a random subset of products on Taobao. We expect the effect magnitude of the PE score to depend on the specific product market. However, we construct the algorithm based on all available TikTok influencer video ads that have spoken words during the data window. Therefore, the algorithm itself can plausibly generalize.

³² If a product has multiple video ads, it is nontrivial how to attribute sales lift to each ad. See Du et al. (2019) for a model of “multitouch attribution.”

³³ We also observe the number of plays for each video. Play volume can be a noisy measure of engagement because it does not capture how much time users actually spend on a video.

³⁴ We compute these two scores as the sum rather than the average of their pixel-level values to reduce their correlation with the PE score while still capturing a video’s level of engagement and product placement. In particular, the PE score is highly correlated ($\rho = 0.85, p < 0.001$) with the product score computed as the average of its pixel-level values and moderately correlated ($\rho = 0.54, p < 0.001$) with the product score computed as the sum; see Schober et al. (2018) for common definitions of high correlation versus moderate correlation. The high correlation may obfuscate subsequent analysis. Nevertheless, the predictive power of the PE score is robust when we compute engagement and product scores as the average of their respective pixel-level values (see Online Appendix H.6).

³⁵ SSIM is a value between zero and one that measures the perceived similarity between two images. It takes additional contextual information, such as luminance and contrast, into account compared with measures, such as Pearson correlation or mean squared error.

³⁶ Analogously, Levitt and Syverson (2008) tested incentive misalignment in the housing market by comparing home sales by agents who sell for others versus themselves. See Villas-Boas (1994) and Wernerfelt et al. (2021) for analyses of the sharing and internalization of advertising agencies, respectively. See Pei and Mayzlin (2022) for a model in which influencers are paid to review products, and see Nistor et al. (2024) for a model of influencers’ dynamic trade-off between growing and monetizing their followers. Our “dual-persona” angle is also related to Yalcin et al. (2020), who highlighted the dual role of influencers as marketers and educators.

³⁷ The sample of 77 video ads has missing observations for average play, price per video ad, and expected CPM. These variables are excluded from the regression to conserve the sample size.

³⁸ There are many missing values in the number of days since the influencer’s first post. To conserve the sample size, unlike other influencer characteristics in Table 2, we do not subsequently include this characteristic as a predictor of sales lift.

³⁹ The two-step approach to analyzing heterogeneous treatment effects is common in statistics (e.g., Cai et al. 2011); see Kent et al. (2018) for a survey of “predictive approaches to heterogeneous treatment effects.” Technically, the standard two-way fixed effect DID model can be problematic when treatment adoption (ad posting in our case) is staggered (e.g., Callaway and Sant’Anna 2021). The two-step approach identifies a treatment effect (sales lift) for each video in a separate sales regression, thus avoiding the staggered-treatment problem.

⁴⁰ In a previously circulated version of the paper, we used 30-day sales and similarly found the PE score to be a positive and significant predictor of sales lift.

⁴¹ If a product misses its 30-day sales revenue on a given day, we replace the missing value with the product’s 30-day sales on the previous observed day so that we can still use the recursive formula to impute daily sales of all following days. However, we do treat daily sales as missing on this given day in subsequent analysis to respect the fact that sales information is missing and possibly so for reasons that we do not observe. Additionally, imputed daily sales may be negative. The results are robust if we set negative imputed daily sales to zero.

⁴² The largest search engine in China, Baidu provides data on keyword-search dynamics on its Baidu Index platform, a service analogous to Google Trends. The Baidu Index has been used in

academic research to control for unobserved market-level interest in various topics (e.g., Jia et al. 2020).

⁴³ The PE score remains a positive and significant ($p < 0.001$) predictor of sales lift if we include influencer and product features (except the gender and product-category indicators) on a logarithmic scale.

⁴⁴ See <https://cran.r-project.org/web/packages/xgboost/vignettes/xgboost.pdf>.

⁴⁵ Larger values of n increase model fit, possibly via overfitting, but they do not change the feature importance metrics much. For example, when $n = 10$, the R^2 and the adjusted R^2 both become 0.93, whereas the PE score remains the most important feature, with a 0.871 gain, 0.240 cover, and 0.280 frequency, not too far from their values in Table 6. In addition, we train an XGBoost model with five-fold crossvalidation and $n = 10$ (the sample of treated products is likely too small to tune hyperparameters). The average R^2 and the adjusted R^2 on the test sets are 0.42 and 0.37, respectively, comparable with the values obtained on the entire treated group at $n = 2$. These results suggest that the feature importance findings are unlikely to be artifacts of overfitting.

⁴⁶ See <https://pjreddie.com/darknet/yolo>.

⁴⁷ See <https://pypi.org/project/fer>.

⁴⁸ See <https://github.com/deepmind/kinetics-i3d>.

⁴⁹ Shorter segments may capture finer dynamics in a video, but they give the algorithm less data to work with in each segment. We also tried segments of 5 or 10 seconds, but the algorithm did not reliably identify activities in these shorter segments.

⁵⁰ The optimal number of topics is selected via Cao et al. (2009) and Deveaud et al. (2014).

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Online Appendix

Engagement that Sells: Influencer Video Advertising on TikTok

Marketing Science

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October 23, 2024

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A 3D Convolutional Neural Network (3D CNN)

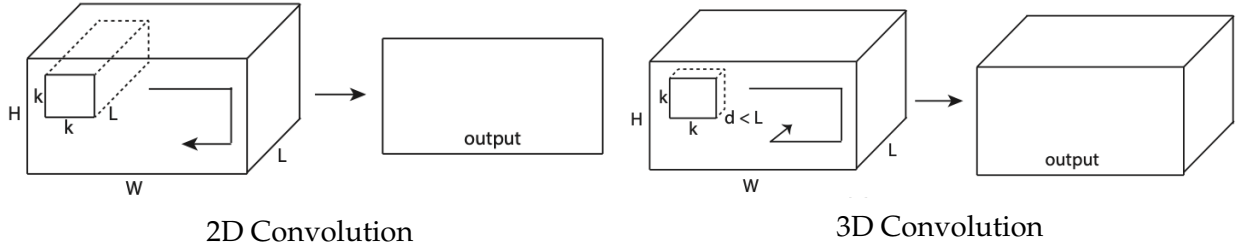
We use a 3D CNN and gradient-based saliency maps to estimate the engagement heat map from observed video-level engagement data (the number of shares, likes, or comments). We use the number of shares as the outcome variable in the main analysis, and verify robustness using the number of likes and comments, respectively. As discussed in the paper, each video in our data is represented as an $(S, 224, 224, 3)$ numerical array, where S is the length of the video in seconds, $(224, 224)$ is the height and width of each video frame in pixels, and 3 is the number of RGB color channels. The output is a single numerical value representing the predicted number of shares of the video. This is a supervised learning problem.

The key building blocks of a CNN are convolution layers. A convolution layer uses filters with weights that are trainable to transform the input images by representing them in a more abstract feature space that captures more general properties of the images (e.g., the presence of an edge or face). Which properties are captured depends on what the network is trained for. Multiple convolution layers can be stacked on top of each other and interspersed with other non-trainable layers such as max pooling layers (to reduce the dimension of feature space), non-linear activation layers (to perform non-linear transformations of input values), and dropout layers (to randomly set some weights to zero to avoid overfitting). After layers of transformation, the feature maps are flattened into a vector and fed into a fully connected layer for the final classification or regression task. We refer interested readers to Saha (2018) for a visual introduction that describes the architecture of a typical CNN and animates what each layer does.

CNNs have been used to analyze images for marketing research in a growing number of papers (e.g., Liu et al. 2020, Hartmann et al. 2021, Zhang et al. 2022, Dzyabura et al. 2023, Tkachenko and Jedidi 2023, Troncoso and Luo 2023, Zhang and Luo 2023). These papers are built on 2D CNNs. In our paper, we use a 3D CNN to account for the additional time dimension of video content. We highlight the difference between a standard 2D

convolution and a 3D convolution in Figure A.1. In a 2D convolution, the filter and the input always have the same depth L , which represents the three color channels. The filter only slides across the spatial dimensions of the input (H and W), which means the output is a 2D matrix. In contrast, the filter in a 3D convolution has a variable depth $d < L$, where L represents the three color channels and time. In addition to sliding across the spatial dimensions, the filter also slides across the depth dimension, outputting a 3D matrix.

Figure A.1. 2D versus 3D Convolution (Tran et al. 2015)



More specifically, we build on Xception (Chollet 2017) pre-trained on ImageNet to extract features from each frame (in a time-distributed manner). Because the top layer of Xception performs a classification task, we remove the top layer while keeping weights in other layers frozen. We stack a 3D convolution layer with 128 units and a filter size of $(3, 3, 3)$ on top of the extracted feature sequence to account for the temporal dependency across frames. We also include a max pooling layer to reduce the dimension of the feature space. The standard max pooling layer outputs a feature map whose dimension depends on the dimension of the input feature map produced by the 3D convolution layer. This approach does not work in our case because our algorithm takes in videos of different lengths. We instead use a global max pooling layer to map variable input feature map dimensions into a fixed dimension. We then add a 128-unit dense layer and a dropout layer (with a dropout rate of 0.1), which has been shown to be particularly effective at reducing overfitting (Srivastava et al. 2014) on top of the global max pooling layer. The final layer is a one-unit dense layer to output the predicted engagement of a video.

The model is optimized with Adadelta¹ against the mean absolute percentage error (MAPE) loss with a learning rate that is initially 0.001 and adjusted adaptively in the training process. The architecture of our network on top of Xception is summarized in Figure A.2.² Hyperparameters such as the number of units in the 3D convolution layer and dense layer, filter size, dropout rate, and initial learning rate are tuned via a grid search on a smaller training sample with 1,000 videos. The optimal combination is chosen based on validation error as detailed below.

Figure A.2. CNN Layers on Top of Xception

Layer (type)	Output Shape	Param #
time_distributed (TimeDistri	(None, None, 7, 7, 2048)	20861480
conv3d (Conv3D)	(None, None, 5, 5, 128)	7078016
global_max_pooling3d (Global	(None, 128)	0
dense (Dense)	(None, 256)	33024
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 1)	257
Total params: 27,972,777		
Trainable params: 7,111,297		
Non-trainable params: 20,861,480		

As explained in the paper, prior to training, we regress raw engagement on product fixed effects, influencer fixed effects, acoustic features, and spoken content. We then normalize the regression residuals to [0, 1] for training. To derive acoustic features, we extract a numerical representation (amplitude) of the sound wave in each video. The raw sampling rate is 44,100 per second. We down-sample it to 100 evenly spaced observations per audio file for tractability. To derive text features, we extract the transcript (mostly in Chinese) of each video using Google Speech-to-Text API.³ We then use a pre-trained

¹<https://keras.io/api/optimizers/adadelta>.

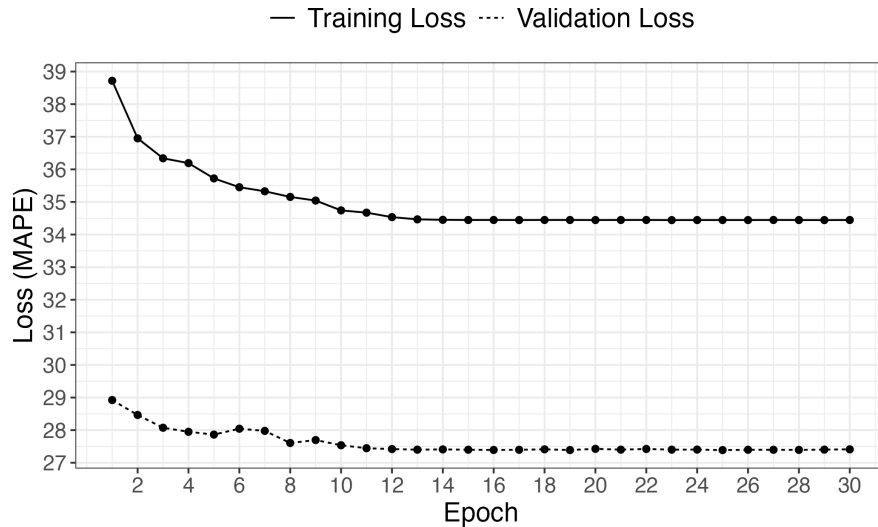
²Xception has 132 layers, hence our full network has 132 + 6 = 138 layers. See Chollet (2017) for more details on the architecture of Xception.

³<https://codelabs.developers.google.com/codelabs/cloud-speech-text-python3>.

multilingual BERT model⁴ to convert the transcripts into 768-dimension embeddings.

Our final 3D CNN has more than 7 million trainable parameters. We train it on 10,000 videos and validate it on 3,500 videos starting from weights optimized on 1,000 videos to speed up convergence. The model is trained on a high-performance computing cluster using TensorFlow (<https://keras.io>). Figure A.3 summarizes the training and validation losses over 30 epochs. The training losses are higher than validation losses because a dropout layer is used in training but not in validation. Both loss curves become flat as the number of epochs increases and do not suggest signs of overfitting (e.g., Salman and Liu 2019). We retain the parameters at the epoch with the minimal validation error as the final model (epoch 25). The accuracy (one minus MAPE) on the holdout test set of 3,451 videos is 73%, which is comparable with recent results on predicting video ad engagement. For example, Chaturvedi et al. (2021) predicted watch time on YouTube video ads with a graph-embedding model and reported an accuracy of 78%.

Figure A.3. Training and Validation Loss



Note: The loss curves are based on the training set of 10,000 videos and the validation set of 3,500 videos. The training losses are higher than the validation losses because a dropout layer is used in training but not validation. Minimal validation loss is achieved at epoch 25.

⁴https://tfhub.dev/tensorflow/bert_multi_cased_L-12_H-768_A-12.

As the main contribution of this paper is the PE score concept, not a new predictive model that achieves higher accuracy, any model that can be used to generate saliency maps (such as CNNs or transformer-based models) can be implemented in the paper’s framework. We used one of the state-of-the-art models (3D CNNs) at the time of this study as a proof of concept and are open to the possibility that other predictive models may enhance the PE score’s efficacy in future applications.

B Engagement Heat Map

B.1 Construction of the Engagement Heat Map

We compute the engagement heat map as a saliency map. A saliency map is a gradient-based visualization method for CNNs (Simonyan et al. 2014). It takes a trained network and computes the gradient of the outcome with respect to an input image. Each entry of the map represents the partial derivative of the outcome with respect to a particular pixel in the input image. Usually, the absolute value of the gradient is used. A high absolute value suggests that a small change in that pixel will lead to a big change in the outcome. For images with color, there are three color channels (RGB). It is typical to compute the gradient for each channel and take the maximum across channels as the final value for that pixel. The eventual output of a saliency map is of the same dimension as the input image, except that the three color channels, as explained, are flattened into one layer.

We adapt the saliency map to videos, which are sequences of images (frames). Instead of computing the gradient with respect to pixels frame by frame, we do so with respect to pixels in the entire video. This allows us to capture any dependency across video frames. More formally, we define pixel-level engagement as:

$$e_{hws} := \max_{\{r,g,b\}} \left(\left| \frac{\partial \hat{f}}{\partial x_{hwsr}} \right|, \left| \frac{\partial \hat{f}}{\partial x_{hws g}} \right|, \left| \frac{\partial \hat{f}}{\partial x_{hws b}} \right| \right)$$

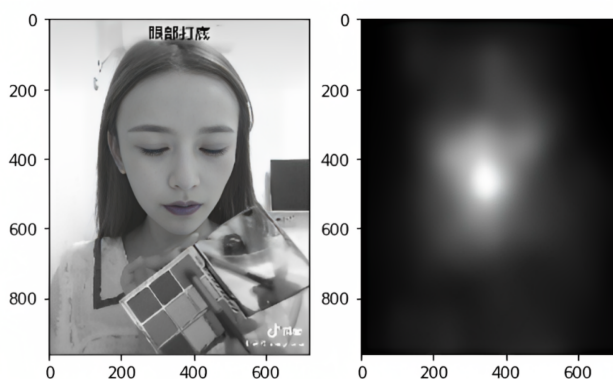
where $\hat{f}$ is the trained 3D CNN, and x_{hwsr} , $x_{hws g}$, and $x_{hws b}$ are the pixel values in the three color channels, respectively, at location (h, w, s) in a video, with h being the index for height in pixels, w for width in pixels, and s for time in seconds. For any input image, these partial derivatives are evaluated at the pixel values of this image.

We use a saliency map to compute pixel-level engagement and call it the (supervised) engagement heat map. It is supervised because the saliency map builds on a 3D CNN trained on video-level engagement data. In Online Appendix H.1, we also discuss an

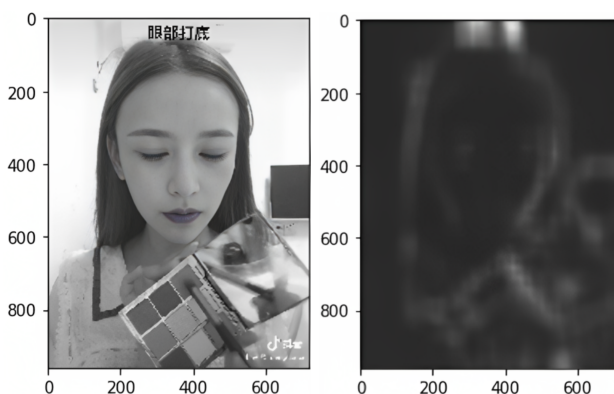
unsupervised approach to engagement heat maps that only requires the video itself.

We implement the supervised saliency map with tf-keras-vis⁵ and the unsupervised saliency map with the saliency module in OpenCV.⁶ Figure B.1a presents an example of a video frame and its corresponding frame in the supervised engagement heat map. Figure B.1b presents an example of the same video frame in the unsupervised engagement heat map.

Figure B.1. An Example of the Engagement Heat Map



(a) Supervised Engagement Heat Map



(b) Unsupervised Engagement Heat Map

Note: The engagement heat map of a video is 3D. We present one frame of this 3D heat map in this figure for illustration. A frame from the example video is shown in the left column. The corresponding frame in the engagement heat map is in the right column (supervised on the top, unsupervised at the bottom). Brighter areas in the engagement heat map correspond to pixels with higher saliency.

⁵<https://github.com/keisen/tf-keras-vis>.

⁶<https://docs.opencv.org/master/d8/d65/group.html>.

B.2 Face-Validity Check of the Engagement Heat Map

As discussed in the paper, the literature has identified the human face as an engaging object that attracts likes and comments on social media (e.g., Bakhshi et al. 2014, Li and Xie 2020, Hartmann et al. 2021). Therefore, as a simple sanity check of the PE-score algorithm, we identify human faces in the videos to see if they are predictive of pixel-level engagement.

We use a face detection algorithm to locate human faces across all frames in a video.⁷ For each frame, the algorithm outputs the location of boxes that contain a human face. Similar to the product heat map, we estimate a face heat map where the values inside the boxes are coded as 1 and values outside are coded as 0. We then compute the correlation between 3D pixel-level engagement values with the indicator variable of whether a face is present in a pixel. The correlation is indeed positive and significant ($\rho = 0.04$, $p < 0.001$). This adds face validity (pun intended) to the engagement heat map because we are now more confident that it is uncovering the relatively engaging parts of a video as we intended.

⁷<https://pypi.org/project/face-recognition>.

C Product Heat Map

We use the scale-invariant feature transform (SIFT) algorithm to detect whether an advertised product appears in a given pixel of the video. We implement SIFT via Oriented FAST and Rotated BRIEF (ORB) in OpenCV.⁸

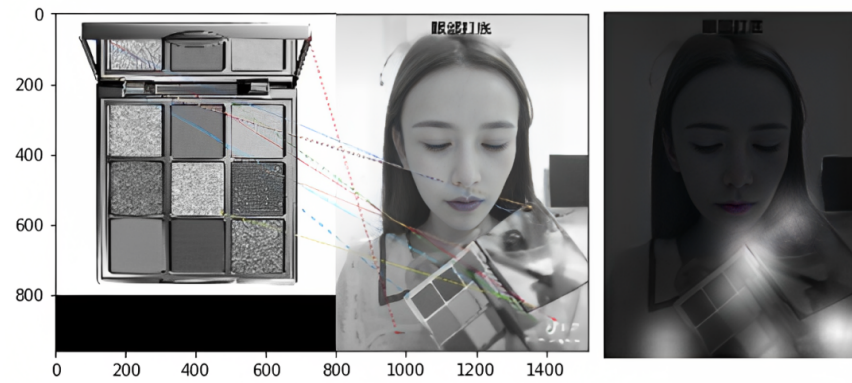
Figure C.1 presents an example. On the left is an image of the advertised product. In the middle is a frame from the video ad. The dashed lines represent connections between the product image and the video frame that we detect using SIFT. These connections indicate the number and location of good keypoint matches. In most cases, the number will not be zero due to noise. A threshold is usually applied to filter out frames with false positive results. In this example, SIFT is able to correctly identify the product from the video despite substantial product rotation.

Following the ratio-test threshold of 0.75 explained in the paper, we assign binary values where 1 indicates that the product is detected at a given pixel and 0 indicates the opposite. On the right of Figure C.1 is the corresponding frame from the product heat map of the video. The bright areas correspond to pixels where SIFT detects product presence. The detected product pixels can be scattered in the frame and do not necessarily enclose the entire product. We create a convex hull of the detected product pixels and consider all pixels within the hull as product pixels.

Similar to the 3D CNN, any model that can be used to detect a specific object can be implemented in the paper’s framework. We used SIFT as a proof of concept and are open to the possibility that other detection models may enhance the PE score’s efficacy in future applications.

⁸https://docs.opencv.org/3.4/d1/d89/tutorial_py_orb.html.

Figure C.1. An Example of the Product Heat Map



Note: We use SIFT to detect the product (left) in a video frame (middle). The corresponding frame in the 3D product heat map of the video is shown on the right, where the bright areas indicate product presence. Random noises are added around detected product pixels to aid visualization.

D Transcript of Practitioner Interviews

To better understand the institutional background of influencer video advertising on TikTok, we interviewed a number of practitioners in the space. We present the interview transcript below (translated into English).

Interviewee: ThinkCrow, TikTok influencer with 1.6 million followers in the science book (lifestyle) category.

- Question: Who determines the content design of the video?

Answer: We have a content design team responsible for this.

- Question: Will advertisers interfere with content design?

Answer: No interference at all.

- Question: How do you determine when to post an advertising video?

Answer: The advertising time does not affect the result very much. There is no special design.

Interviewee: Yuerong Zhao, Senior Project Manager of a TikTok influencer incubation company that has more than 30 influencers in the household and makeup categories.

- Question: Who determines the content design of the video?

Answer: Our content design team.

- Question: Will advertisers interfere with content design?

Answer: It depends on the strength and popularity of the brand [advertiser]. Powerful brands may interfere a little, but not too much. [When further probed:] They will not interfere with the position of the products in the video.

- Question: How do you determine when to post an advertising video?

Answer: There is not much planning for the ad posting time. Sometimes, ads will be posted before the Double 11 Festival. In most cases, there is no specific time.

- Question: What other advertising channels do your customers [advertisers] have?

Answer: We have an exclusive agreement not to release the ad elsewhere.

Interviewee: Name undisclosed, the person in charge of TikTok e-commerce live broadcast products in all categories.

- Question: Will advertisers interfere with content design?

Answer: Sometimes the influencer is asked to speak for a specific amount of time.

Sometimes there may be materials suggested for the influencer to use.

Interviewee: Jian Qin, Senior Product Manager of TikTok.

- Question: Will advertisers interfere with content design?

Answer: Generally, influencer companies have scripts or video samples taken in the past. Some advertisers will also provide the advertising language and scripts, all text, they want to display.

Interviewee: Lei Zhou, Xingtu Advertising Business Affiliate.

- Question: Will advertisers interfere with content design?

Answer: If the advertiser's company is very small and they want to spend less money, they won't care much about how the content is designed. If it is a large business, the advertiser will review the video to make sure there is no text content that damages the brand image. [When further probed:] However, advertisers generally do not interfere with the location and time of the product placement and the video production method. Strong influencers are hardly interfered with by advertisers.

In summary, these interviews suggest that (1) sellers do not tend to influence the visual aspect of video content that we focus on in the paper, (2) sellers do not tend to influence product placement in the video ad, and (3) influencers do not tend to choose the posting time of video ads based on product-specific demand.

E Predicting Missing Product Category Information

Among the products in the sales panel, 68% are missing category labels. To be able to include product category information in the analysis, we predict missing category labels based on product titles. To do so, we draw on a sample of 8,447 products with category labels (including products outside the sales panel to increase the sample size). We assign 70% of products in this sample into the training-and-validation set and perform 10-fold cross-validation. We keep the remaining 30% as the holdout test set. We also make sure that the ratio of training-and-validation to test data in each category is 70:30.

For pre-processing, we use the packages `quanteda`,⁹ `stopwords`,¹⁰ and `chinese.misc`¹¹ to tokenize the titles, delete stop words, and keep the nouns only. For feature extraction, we first construct a term-document matrix. Next, because titles from the same category often share common words, we use latent semantic analysis (LSA), which measures word-word, word-passage, passage-passage relations by applying singular value decomposition (SVD) to factorize the term-document matrix. Finally, we train the model with `XGBoost`¹² in `Caret`.¹³ The model achieves 82% accuracy in the test sample. We have also tried `ranger` and `rpart`, which achieve 63% and 79% accuracy, respectively. Based on predictive accuracy, we use the trained `XGBoost` model to impute missing category labels for products in the sales panel.

⁹<https://cran.r-project.org/web/packages/quanteda/quanteda.pdf>.

¹⁰<https://cran.r-project.org/web/packages/stopwords/stopwords.pdf>.

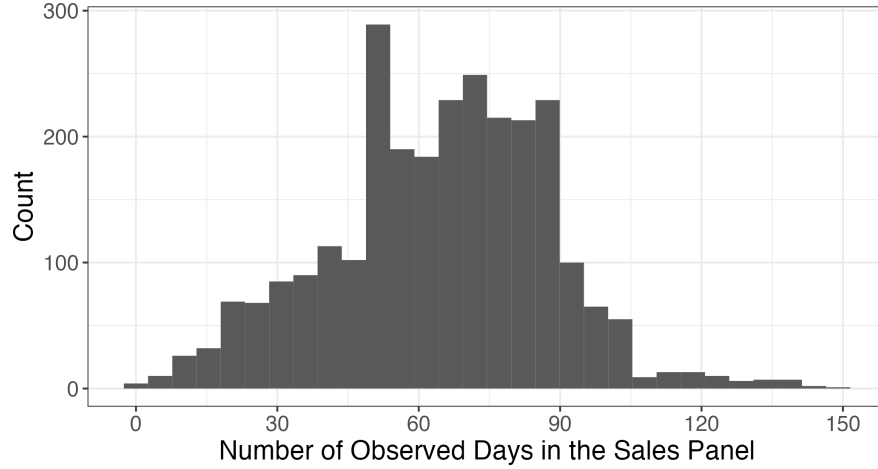
¹¹<https://cran.r-project.org/web/packages/chinese.misc/chinese.misc.pdf>.

¹²<https://cran.r-project.org/web/packages/xgboost/vignettes/xgboost.pdf>.

¹³<https://cran.r-project.org/web/packages/caret/caret.pdf>.

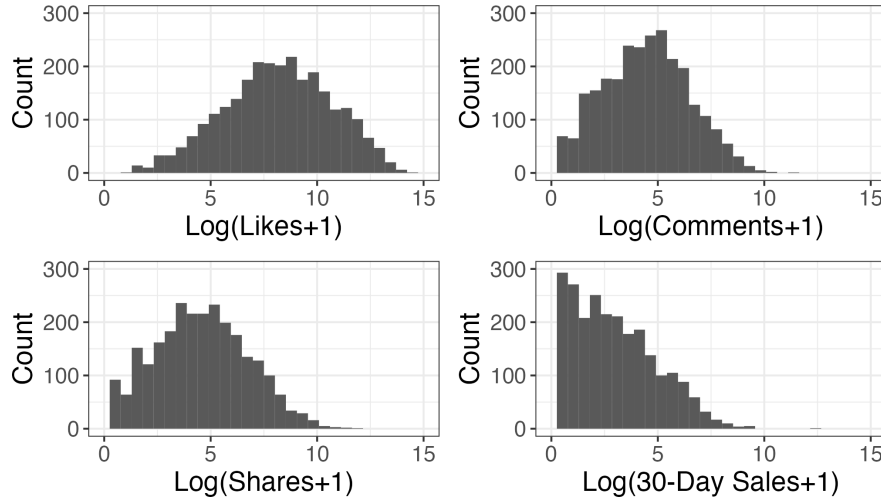
F Additional Summary Statistics of the Sales Panel

Figure F.1. Distribution of the Number of Observed Days in the Sales Panel



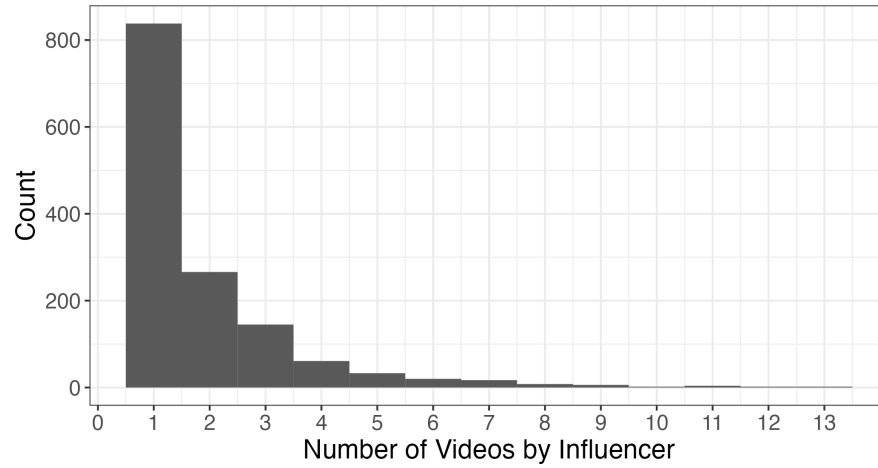
Note: The sample consists of all products in the sales panel. Each observation is a product.

Figure F.2. Distribution of Observed Video Engagement and Product Sales



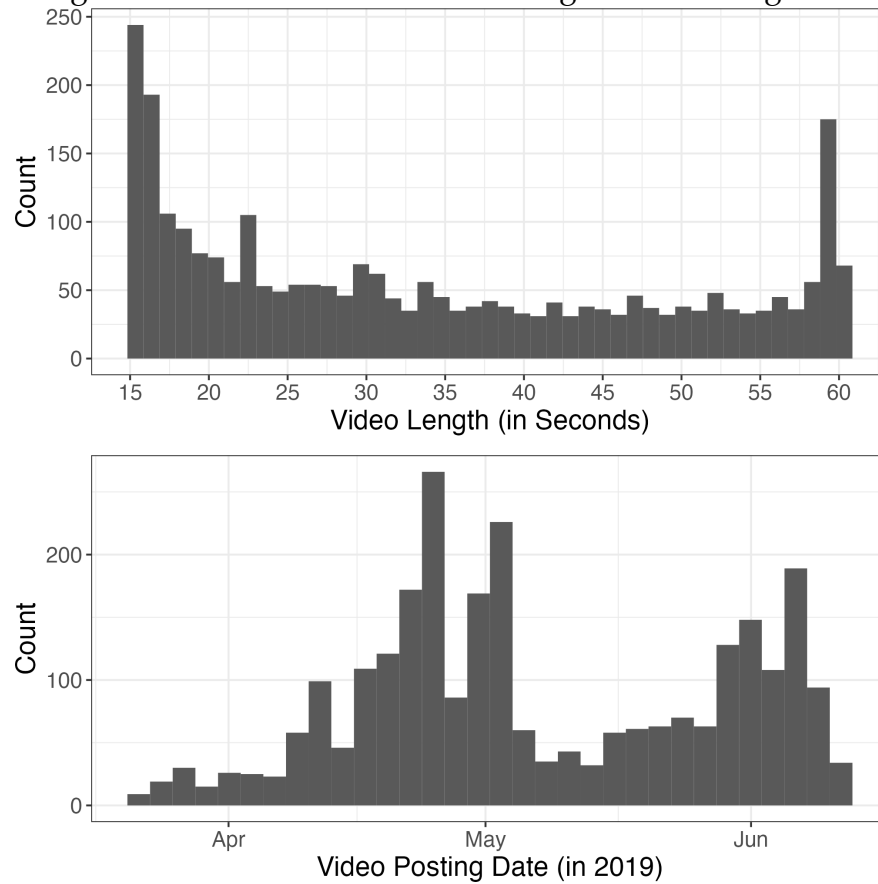
Note: The sample consists of all videos/products in the sales panel, where each product corresponds to one video ad. The subfigures present, in order, the distribution of the video-level number of likes, comments, and shares, and the product-level average 30-day sales revenue in the sales panel as defined in the paper, all on a logarithmic scale to facilitate visualization.

Figure F.3. Distribution of the Number of Videos by Influencer

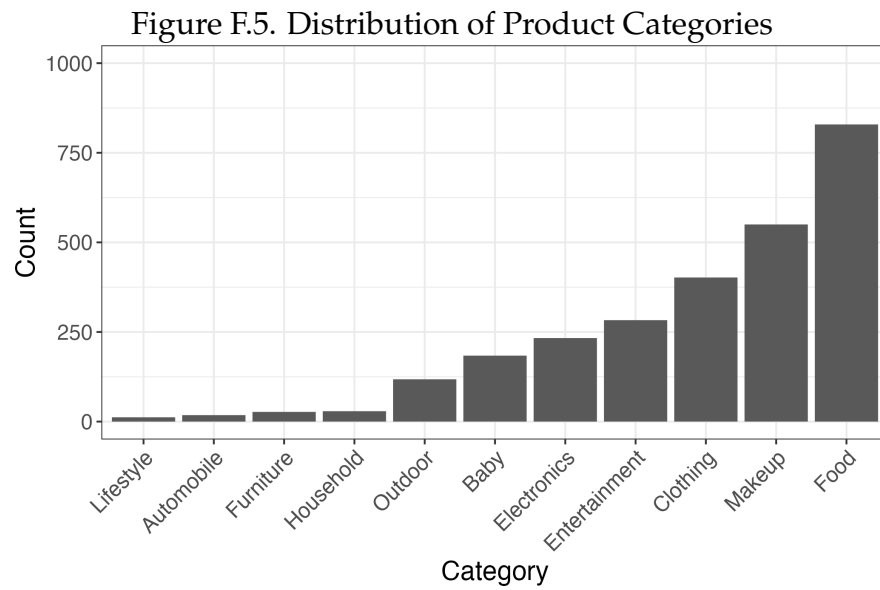


Note: The sample consists of all influencers in the sales panel. Each observation is an influencer.

Figure F.4. Distribution of Video Length and Posting Time

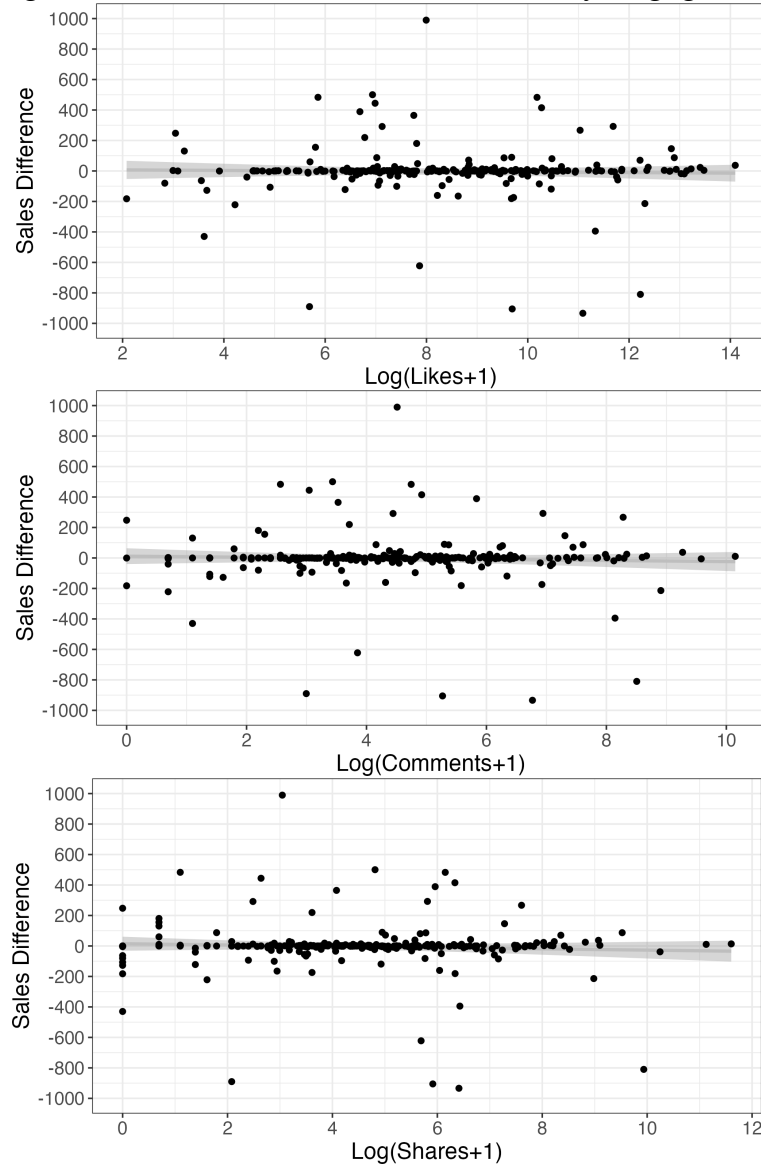


Note: The sample consists of all videos in the sales panel. Each observation is a video.



Note: The sample consists of all products in the sales panel. Product-category indicators will be included as predictors of sales lift.

Figure F.6. Before-After Sales Difference by Engagement



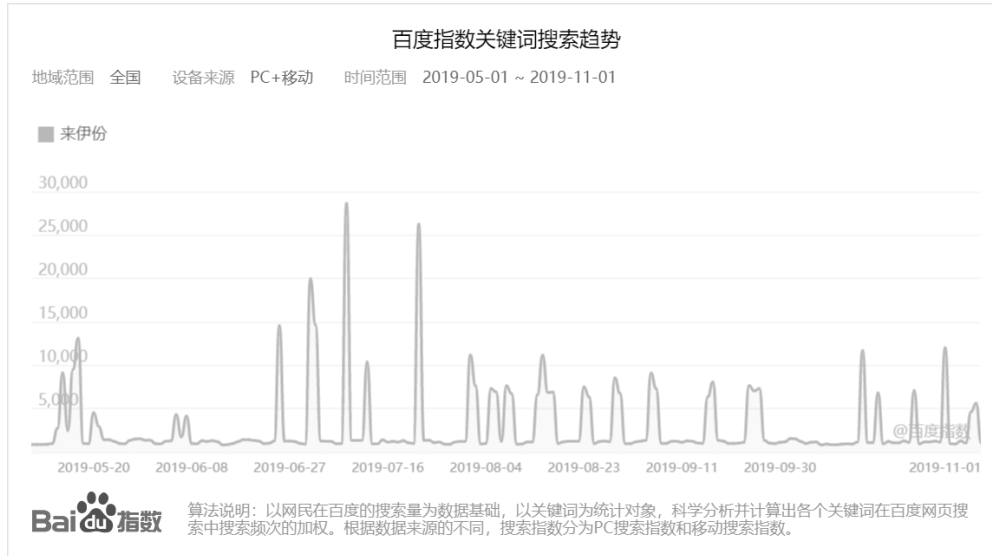
Note: The sample consists of the 259 products in the sales panel that had a video ad posted during the data window. The x-axis is a product's raw video-engagement metric (the number of likes, comments, and shares, respectively) on a logarithmic scale to control for outliers. The y-axis is a product's sales difference, which equals the product's average 30-day sales revenue (in 1,000 RMB) after posting its video ad minus its average 30-day sales revenue before. We restrict the y-axis to values between $-1,000$ and $1,000$ for visualization and the resulting figure displays over 97% of the treated products. The gray areas represent the 95% confidence band along the regression line. Over these 259 products, the correlation between $\log(\text{engagement}+1)$ and sales difference is 0.00025 ($p = 0.997$) for likes, -0.00082 ($p = 0.99$) for comments, and 0.047 ($p = 0.45$) for shares, respectively; the linear regression of sales difference on $\log(\text{engagement}+1)$ and an intercept yields a coefficient of 0.10 ($p = 0.997$) for likes, -0.40 ($p = 0.99$) for comments, and 20.05 ($p = 0.45$) for shares, respectively. The correlation between raw video engagement and sales difference is -0.0075 ($p = 0.90$) for likes, -0.020 ($p = 0.74$) for comments, and -0.0074 ($p = 0.91$) for shares, respectively; the linear regression of sales difference on raw engagement and an intercept yields a coefficient of -0.000055 ($p = 0.90$) for likes, -0.0093 ($p = 0.74$) for comments, and -0.00087 ($p = 0.91$) for shares, respectively.

G Baidu Search Index

As a proxy of unobserved time-varying demand, we collected data on the Baidu search index for all 2,685 products in the sales panel. Two research assistants manually entered the brand of each product as the keyword to track on the Baidu Index website (batch data collection is not available). The Baidu Index website currently accommodates keyword searches at the level of product brand, not specific products. Nevertheless, we expect the search results to capture unobserved demand shifters such as brand campaigns or product campaigns that generate spillover effects within the same brand.

The scope of a keyword search was set to include queries from all over the country (China), from both personal computers and mobile devices, and from May 1 through November 1, 2019, to match the time frame of the sales panel. Figure G.1 presents an example of search results from one keyword.

Figure G.1. Baidu Search Index Screenshot



Out of all the products in the sales panel, we were able to obtain Baidu search results for 429 products. Visual inspection suggests that these tend to be products from bigger, more recognizable brands. For products without search results, we treat their search data as sequences of zeros. Replacing zeros with other constants does not affect the identification of sales lift because we include product/video fixed effects in the analysis.

H Robustness Checks

In this section, we extend the main analysis presented in the paper to check the robustness of the PE-score algorithm. We check three aspects of robustness, with respect to the construction of the algorithm, the measurement of sales lift, and the predictors of sales lift.

For all robustness checks, we test the predictive power of the PE score in both the OLS and XGBoost frameworks. We report robustness-check results when we rerun column (6) of Table 5 (the OLS specification with the most features) and Table 6 (the table of the most important features in XGBoost), unless otherwise noted.

H.1 Alternative Construction of the Engagement Heat Map

In the main analysis, we use the number of shares as the outcome variable to train the 3D CNN and extract saliency maps. As a first robustness check, we retrain the algorithm using the number of likes and comments instead and rerun all subsequent analysis that relies on the PE score as well as the engagement score. (The identification of sales lift is not affected.) Columns (1) and (2) of Table H.1 present the OLS estimation results. The PE score is a positive and the only significant predictor of sales lift at $p < 0.05$ in both alternative scenarios. The top and middle portions of Table H.2 present the most important features from the XGBoost model when the algorithm is trained on the number of likes and comments, respectively. In both cases, the PE score continues to be the most important predictor of sales lift by far according to gain.

So far, we have used the supervised approach to construct the engagement heat map, drawing on video content as input and observed video-level engagement (the number of shares, likes, or comments) as output. Pixel-level engagement is thus determined in a supervised way; a pixel will have a high engagement score if a small change in its value affects observed video-level engagement by a large amount. We also check if the algorithm works when engagement is computed in an unsupervised approach.

Table H.1. Predicting Sales Lift
(Alternative Construction of the Engagement Heat Map)

		Dependent Variable: Sales Lift		
		(1)	(2)	(3)
		Constructed on # Likes	Constructed on # Comments	Constructed Unsupervised
Computed Scores	PE Score	119.36* (53.21)	164.78* (66.83)	12.20 (30.88)
	Engagement Score	−5.35 (22.84)	−6.26 (27.10)	−5.92 (31.18)
	Product Score	−26.76 (57.17)	29.48 (59.39)	49.92 (63.98)
	Engagement Score × Product Score	31.14 (98.74)	−61.26 (111.44)	−90.94 (118.93)
Influencer Features	Gender	5.86 (6.48)	5.89 (6.46)	7.56 (6.46)
	# Followers (in Millions)	0.62 (1.52)	0.34 (1.52)	0.58 (1.54)
	Average Play (in Millions)	0.64 (2.38)	0.45 (2.36)	0.22 (2.40)
	Price per Video Ad (in Millions)	−72.73 (185.76)	−49.75 (186.36)	−71.10 (188.99)
	Expected CPM (in Millions)	9.87 (44.60)	9.93 (44.46)	5.36 (44.86)
	# Video Ads Influencer Has Posted	−0.01 (0.18)	−0.003 (0.18)	−0.02 (0.18)
Product Features	Average Search	−3.28 (7.23)	−3.48 (7.24)	−4.40 (7.27)
	Price	0.002 (0.01)	0.002 (0.01)	0.002 (0.01)
	Discount	−0.003 (0.005)	−0.003 (0.005)	−0.003 (0.005)
	Product-Category Indicators	Yes	Yes	Yes
	Observations	259	259	259
R ²		0.05	0.06	0.04
Adjusted R ²		−0.04	−0.03	−0.05

Note: The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level, or video-level, sales lift in 1,000 RMB. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The motivation for the unsupervised approach is that engagement may be correlated with intrinsic properties of the images themselves. Salient regions in an image may disproportionately affect overall engagement. Past research shows that, indeed, saliency measures based on intrinsic properties of images predict actual gaze and eye movement (e.g., Itti 2005, Dupont et al. 2016). Moreover, the unsupervised approach does not rely on

Table H.2. XGBoost Feature Importance in Predicting Sales Lift
(Alternative Construction of the Engagement Heat Map)

Feature	Gain	Cover	Frequency
Engagement Heat Map Constructed on # Likes			
PE Score	0.876	0.256	0.318
Discount	0.043	0.188	0.205
Product Score	0.041	0.370	0.205
# Followers	0.020	0.139	0.091
# Video Ads Influencer Has Posted	0.013	0.019	0.045
Average Search	0.006	0.011	0.023
Price	0.001	0.009	0.023
Gender	0.0004	0.002	0.023
Engagement Score	0.0003	0.003	0.023
Expected CPM	0.0002	0.002	0.023
Engagement Heat Map Constructed on # Comments			
PE Score	0.908	0.184	0.364
Product Score	0.030	0.171	0.152
Engagement Score	0.017	0.167	0.121
Discount	0.015	0.016	0.091
# Video Ads Influencer Has Posted	0.010	0.153	0.061
# Followers	0.007	0.002	0.030
Average Search	0.006	0.075	0.030
Engagement Score $\times$ Product Score	0.004	0.082	0.061
Product Category: Clothing	0.003	0.070	0.030
Expected CPM	0.002	0.076	0.030
Unsupervised Engagement Heat Map			
Average Play	0.863	0.208	0.194
Expected CPM	0.057	0.165	0.161
Product Score	0.028	0.154	0.129
Engagement Score	0.026	0.147	0.097
PE Score	0.017	0.282	0.194
# Followers	0.010	0.002	0.065
Discount	0.0004	0.018	0.065
Average Search	0.0002	0.017	0.065
Product Category: Makeup	0.0001	0.008	0.032

Note: The sample for each portion of the table consists of all treated products in the sales panel. The features in each portion are ranked by gain, the standard metric of feature importance for the XGBoost model's predictive accuracy. Each portion of the table reports up to 10 of the most important features of the XGBoost model trained on its entire sample. When the number of features shown in a portion of the table is less than 10, the unshown features are not used by the XGBoost model to split nodes, meaning they are unimportant according to the model.

video engagement data; pixel-level engagement is determined by the images themselves.

To construct the unsupervised engagement heat map, we use the intrinsic properties of the images (the statistically distinct areas of an image, such as high-contrast locations and edges of objects; see Figure B.1b for an example) as a proxy for pixel-level engagement (Hou and Zhang 2007). As column (3) of Table H.1 shows, the PE score based on unsupervised learning does not predict sales lift with statistical significance. As the bottom portion of Table H.2 shows, the PE score is no longer the most important predictor of sales lift based on gain. These results suggest that statistically distinct areas do not imply higher engagement in the videos we study. There is substantial value in collecting video engagement data to construct a supervised engagement heat map.

H.2 Alternative Control Products and Treatment Specification

We revisit the causal identification of sales lift in this section. When the treatment effect varies over time, it may be problematic to use products that have already been treated as controls (Callaway and Sant’Anna 2021). We address this potential concern with two alternative ways to define control products.

First, we only include in the control group products that did not post influencer video ads during the 30 days before they entered the sales panel. The reason is that any impact of the video ad on the product’s daily sales trajectory is likely to have diminished given that video popularity tends to be short-lived on today’s social media platforms. For context, Huang and Morozov (2024) analyzed video advertising by Twitch influencers and found an hourly ad carryover coefficient of 0.712, which is equivalent to a daily carryover coefficient of only 0.0003. Excluding products that posted video ads within the prior 30-day window leaves us with 357 control products.

Second, we focus on products not yet treated and use their pre-treatment sales trajectories as controls. Specifically, we select products that posted their video ads after June 1, 2019 and use their sales panel before their respective ad posting dates as controls. This splits the 259 treated products into 113 control products and 146 treated products, among which 140 treated products have corresponding controls and thus identified sales lift.

Under both alternative definitions of control products, the average sales lift remains insignificantly different from zero ($p = 0.60$ and $p = 0.19$, respectively). Columns (1) and (2) of Table H.3 present the OLS estimation results, where the PE score remains a positive predictor and the only significant predictor of sales lift ($p < 0.001$) under both alternative definitions of control products. The top and middle portions of Table H.4 report the XGBoost feature importance results under these two definitions of control products. The PE score remains the most important predictor of sales lift based on gain. Notably, the predictive power of the PE score remains distinctively strong despite the much smaller number of control products under both alternative definitions of the control group.

Table H.3. Predicting Sales Lift
(Alternative Definitions of Control Products and Treatment Specification)

		Dependent Variable: Sales Lift		
		(1)	(2)	(3)
		Products without Recent Ads as Control Group	Products Not Yet Treated as Control Group	Alternative Treatment Specification
Computed Scores	PE Score	102.59*** (29.12)	511.25*** (141.52)	1,737.00** (522.41)
	Engagement Score	−2.97 (32.60)	148.79 (157.06)	438.64 (567.62)
	Product Score	−109.94 (81.04)	−207.46 (390.44)	−649.23 (1,382.84)
	Engagement Score × Product Score	113.34 (158.02)	−146.20 (751.80)	−697.25 (2,692.97)
Influencer Features	Gender	5.74 (6.41)	1.55 (29.22)	31.95 (116.21)
	# Followers (in Millions)	0.18 (1.50)	−3.89 (5.82)	−10.13 (23.87)
	Average Play (in Millions)	0.08 (2.34)	4.97 (11.33)	16.78 (51.47)
	Price per Video Ad (in Millions)	−11.57 (184.97)	623.91 (838.31)	1,400.52 (3,207.84)
	Expected CPM (in Millions)	15.08 (44.16)	33.00 (146.78)	63.05 (680.53)
	# Video Ads Influencer Has Posted	0.01 (0.18)	−0.67 (0.98)	−0.91 (3.22)
Product Features	Average Search	−3.44 (7.15)	−8.24 (31.39)	−11.32 (118.47)
	Price	0.003 (0.01)	0.04 (0.07)	0.17 (0.31)
	Discount	−0.002 (0.005)	−0.05 (0.07)	−0.21 (0.31)
	Product-Category Indicators	Yes	Yes	Yes
	Observations	259	140	191
R ²		0.08	0.15	0.11
Adjusted R ²		−0.003	−0.01	−0.01

Note: The sample for column (1) consists of all treated products in the sales panel. The sample for column (2) consists of all treated products in the sales panel that posted their video ads before June 1, 2019 and that have identified sales lift. The sample for column (3) consists of all treated products in the sales panel that have identified sales lift under the alternative treatment specification. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level, or video-level, sales lift in 1,000 RMB. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table H.4. XGBoost Feature Importance in Predicting Sales Lift
(Alternative Definitions of Control Products and Treatment Specification)

Feature	Gain	Cover	Frequency
Products without Recent Ads as Control Group			
PE Score	0.896	0.273	0.344
Product Score	0.037	0.388	0.219
Expected CPM	0.036	0.091	0.094
Engagement Score	0.020	0.012	0.188
Discount	0.004	0.084	0.062
# Followers	0.004	0.142	0.062
Product Category: Electronics	0.003	0.010	0.031
Products Not Yet Treated as Control Group			
PE Score	0.991	0.185	0.192
Discount	0.003	0.295	0.269
Product Score	0.002	0.171	0.077
Engagement Score	0.001	0.063	0.154
Engagement Score $\times$ Product Score	0.001	0.069	0.077
# Followers	0.001	0.010	0.038
Price per Video Ad	0.001	0.117	0.115
Average Play	0.001	0.069	0.038
Price	0.0001	0.020	0.038
Alternative Treatment Specification (as 1 Only on the Day of Treatment)			
PE Score	0.996	0.245	0.282
# Video Ads Influencer Has Posted	0.002	0.089	0.103
Product Score	0.001	0.229	0.103
Discount	0.0004	0.104	0.103
# Followers	0.0003	0.009	0.026
Product Category: Clothing	0.0003	0.168	0.051
Engagement Score $\times$ Product Score	0.0002	0.008	0.026
Engagement Score	0.0001	0.017	0.103
Expected CPM	0.00003	0.059	0.103
Price per Video Ad	0.00003	0.059	0.077

Note: The sample for the top portion of the table consists of all treated products in the sales panel. The sample for the middle portion of the table consists of all treated products in the sales panel that posted their video ads before June 1, 2019 and that have identified sales lift. The sample for the bottom portion of the table consists of all treated products in the sales panel that have identified sales lift under the alternative treatment specification. The features in each portion are ranked by gain, the standard metric of feature importance for the XGBoost model's predictive accuracy. Each portion of the table reports up to 10 of the most important features of the XGBoost model trained on its entire sample. When the number of features shown in a portion of the table is less than 10, the unshown features are not used by the XGBoost model to split nodes, meaning they are unimportant according to the model.

We examine another causal identification question in this section. The paper specifies the treatment effect as $Post_{vd}$ being 1 if video ad v is posted by day d and 0 otherwise. This treatment effect thus captures the video ad’s average daily sales lift since its posting. An alternative treatment specification is that $Post_{vd}$ equals 1 if video ad v is posted *on* day d and 0 otherwise. This alternative specification suits markets where most of the sales lift occurs on the day of ad posting (e.g., Huang and Morozov 2024).

Under this alternative treatment specification, a treated product relies on the existence of sales records in the control group *on* its treatment day to identify its treatment effect. This condition is not satisfied for all treated products because, as explained in the paper, different products have different numbers of observed days in the sales panel (see Figure F.1 of the Online Appendix). This reduces the effective number of treated products to 191. (Under the original treatment specification, a treated product relies on the existence of sales records in the control group *since* its treatment day to identify its treatment effect, a condition that is easier to satisfy.)

Under the alternative treatment specification, the average sales lift is insignificantly different from zero ($p = 0.32$). Column (3) of Table H.3 presents the OLS estimation results, where the PE score is a positive predictor and the only significant predictor of sales lift ($p < 0.01$). The bottom portion of Table H.4 reports the XGBoost feature importance results, where the PE score is again notably the most important predictor of sales lift based on gain.

One caveat to note is that, if the true ad effect extends beyond the first day, this alternative specification may overestimate baseline product sales, as captured in the product fixed effect, and may in turn underestimate the treatment effect on the day the ad is posted. As such, this alternative specification should be interpreted with caution. Nevertheless, the results suggest that the PE score may continue to predict sales lift if the true environment is such that the ad effect is concentrated on the first day.

Last but not least, if the truth is that ad effect decays over time, the treatment effect

in the main analysis (i.e., average daily sales lift since ad posting) may be mechanically larger for ads posted later in the observation window, other things being equal. To address this possibility, we re-estimate the OLS and XGBoost models, including ad posting date as a control variable. The OLS results are similar to their counterparts in Table 5. The PE score remains the most important feature in the XGBoost model, with a gain of 0.858, comparable to its counterpart of 0.865 in Table 6. Ad posting date is not among the 10 most important features in the XGBoost model.

H.3 Alternative Imputation of Daily Sales

As explained in the paper, we only observe each product’s 30-day sales revenue, based on which we impute the product’s daily sales revenue. Again, consider a representative product v in the sales panel and drop the product subscript v for now. For the main analysis, we assume $Daily\ Sales_t = 30\text{-Day}\ Sales_1/30$, where $t = 1$ denotes the first day the product is observed in the sales panel and $t \in \{-28, -27, \dots, -1, 0, 1\}$ denotes days in its prior 30-day “initialization period.” This equal-allocation rule is based on managerial experience that sales tend to be stable absent promotional events. In this section, we examine three alternative ways to impute daily sales.

First, as discussed in the paper, we collected Baidu search index data for each product in the sales panel to control for its unobserved demand dynamics at the daily level. Making use of this data, instead of allocating $30\text{-Day}\ Sales_1$ evenly over the initialization period, we allocate it in proportion to the product’s daily search volume: $Daily\ Sales_t = 30\text{-Day}\ Sales_1 \times Search_t / \sum_{\tau=-28}^1 Search_{\tau}$. When a product has missing observations of search volume during the initialization period, we replace the missing value with this product’s average search volume.

Second, we exclude all products in the sales panel that have “unusual fluctuations” in search volume, defined as variations beyond one standard deviation of the product’s average search volume, during the initialization period. This leaves us with 243 treated products and 2,214 control products.

Third, we use the Denton-Cholette method (Dagum and Cholette 2006) to disaggregate each product’s 30-day sales time series into a daily sales time series. The Denton-Cholette method is a “purely mathematical” way to disaggregate a lower-frequency time series (e.g., 30-day sales) into a higher-frequency time series (e.g., daily sales) based on the principle of movement preservation. The method does not require a higher-frequency “indicator series” (e.g., daily search volume) for imputation; without an indicator series, it disaggregates the lower-frequency series smoothly. This allows us to check the robust-

ness of the results from a different angle, one that does not rely on search volume, but does ignore the daily sales fluctuations otherwise indicated by search volume. For implementation, we sample each product’s 30-day sales revenue on its first observed day in the data, the 31st day, and so on. We then disaggregate this time series into the product’s daily sales time series using Denton-Cholette.¹⁴

Under all three alternative imputation strategies, the average sales lift remains insignificantly different from zero ($p = 0.50, 0.46$, and 0.90 , respectively). Table H.5 presents the OLS estimation results of the test of interest, where the PE score remains a positive and significant predictor of sales lift ($p < 0.001$ under the first two imputation strategies, and $p < 0.01$ under the third). The top, middle, and bottom portions of Table H.6 report the XGBoost feature importance results under these three imputation strategies, respectively. The PE score remains the most important predictor of sales lift based on gain. Quantitatively, the PE score is less predictive of sales lift under Denton-Cholette than under the other imputation methods. One possible reason is the temporal smoothing effect of Denton-Cholette without an indicator series.

¹⁴We use the “fast” method in the `tempdisagg` package in R, which approximates but is much faster than the Denton-Cholette method with $h = 1$. The latter method minimizes the sum of squares of the deviations between the first differences of the indicator series (none, thus set to a constant of 1 in our application) and the imputed series.

Table H.5. Predicting Sales Lift
(Alternative Imputation of Daily Sales)

		Dependent Variable: Sales Lift		
		(1)	(2)	(3)
		Initializing Daily Sales in Proportion to Search Volume	Removing Products with Unusual Initial Search	Denton- Cholette Temporal Disaggregation
Computed Scores	PE Score	100.79*** (28.92)	104.33*** (30.01)	55.08** (16.77)
	Engagement Score	−2.46 (32.38)	4.13 (34.36)	3.32 (18.77)
	Product Score	−112.44 (80.50)	−112.65 (84.75)	−80.99 (46.66)
	Engagement Score × Product Score	126.29 (156.96)	119.69 (167.98)	67.02 (90.98)
Influencer Features	Gender	5.89 (6.37)	5.54 (6.72)	−0.50 (3.69)
	# Followers (in Millions)	0.24 (1.49)	−0.65 (1.62)	0.23 (0.86)
	Average Play (in Millions)	0.30 (2.32)	1.08 (2.43)	0.62 (1.35)
	Price per Video Ad (in Millions)	−9.38 (183.72)	166.83 (210.77)	−82.13 (106.49)
	Expected CPM (in Millions)	13.09 (43.86)	9.38 (44.89)	8.56 (25.43)
	# Video Ads Influencer Has Posted	−0.01 (0.18)	−0.36 (0.26)	0.21* (0.10)
Product Features	Average Search	−3.69 (7.11)	−12.73 (9.80)	−0.29 (4.12)
	Price	0.003 (0.01)	0.003 (0.01)	−0.0003 (0.003)
	Discount	−0.002 (0.005)	−0.003 (0.005)	−0.001 (0.003)
	Product-Category Indicators	Yes	Yes	Yes
Observations		259	243	259
R ²		0.08	0.10	0.09
Adjusted R ²		−0.005	0.01	0.01

Note: The sample for columns (1) and (3) consists of all treated products in the sales panel. The sample for column (2) consists of all treated products in the sales panel except products with unusual search fluctuations during the initialization period. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level, or video-level, sales lift in 1,000 RMB. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table H.6. XGBoost Feature Importance in Predicting Sales Lift
(Alternative Imputation of Daily Sales)

Feature	Gain	Cover	Frequency
Initializing Daily Sales in Proportion to Search Volume			
PE Score	0.865	0.338	0.242
Expected CPM	0.044	0.075	0.091
Product Score	0.027	0.329	0.152
Engagement Score	0.025	0.006	0.091
Discount	0.017	0.034	0.212
# Followers	0.013	0.070	0.091
Average Search	0.009	0.143	0.061
Price	0.0002	0.002	0.030
# Video Ads Influencer Has Posted	0.0001	0.003	0.030
Removing Products with Unusual Initial Search			
PE Score	0.888	0.174	0.333
Engagement Score	0.035	0.009	0.143
# Video Ads Influencer Has Posted	0.028	0.168	0.095
Product Score	0.019	0.248	0.143
Average Search	0.013	0.163	0.095
Engagement Score $\times$ Product Score	0.007	0.158	0.095
Product Category: Food	0.006	0.003	0.048
Product Category: Clothing	0.003	0.078	0.048
Denton-Cholette Temporal Disaggregation			
PE Score	0.590	0.247	0.379
Product Score	0.287	0.175	0.138
# Video Ads Influencer Has Posted	0.052	0.168	0.069
# Followers	0.037	0.008	0.069
Expected CPM	0.012	0.009	0.034
Average Search	0.011	0.294	0.138
Discount	0.006	0.009	0.069
Product Category: Clothing	0.002	0.078	0.034
Engagement Score $\times$ Product Score	0.002	0.008	0.034
Engagement Score	0.001	0.004	0.034

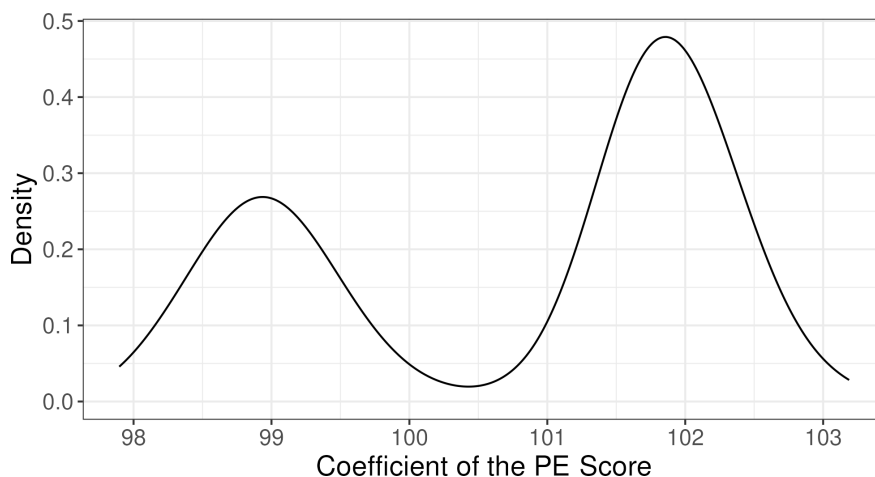
Note: The sample for the top and bottom portion of the table consists of all treated products in the sales panel. The sample for the middle portion of the table consists of all treated products in the sales panel except products with unusual search fluctuations during the initialization period. The features in each portion are ranked by gain, the standard metric of feature importance for the XGBoost model's predictive accuracy. Each portion of the table reports up to 10 of the most important features of the XGBoost model trained on its entire sample. When the number of features shown in a portion of the table is less than 10, the unshown features are not used by the XGBoost model to split nodes, meaning they are unimportant according to the model.

H.4 Capturing Measurement Uncertainty in Sales Lift

One fact to note about the two-step algorithm evaluation approach is that sales lift is estimated with uncertainty.¹⁵ We use bootstrapping to examine the impact of measurement uncertainty in sales lift on the PE score's predictive power (Efron 1992). We construct 1,000 bootstrap samples of the control products. Each sample is based on a random draw of 2,426 products with replacement from the 2,426 control products. For each bootstrap sample, we repeat the process of estimating sales lift of each treated product and then predicting sales lift with OLS and XGBoost, respectively. This generates a distribution of the OLS coefficient of the PE score, its p -value, and its XGBoost feature importance across the 1,000 samples.¹⁶

Figures H.1, H.2, and H.3 below present these distributions. In all 1,000 bootstrap samples, the PE score is a positive and significant ($p < 0.001$) predictor of sales lift in the OLS model and the most important predictor in the XGBoost model based on gain.

Figure H.1. Distribution of the Coefficient of the PE Score Across Bootstrap Samples

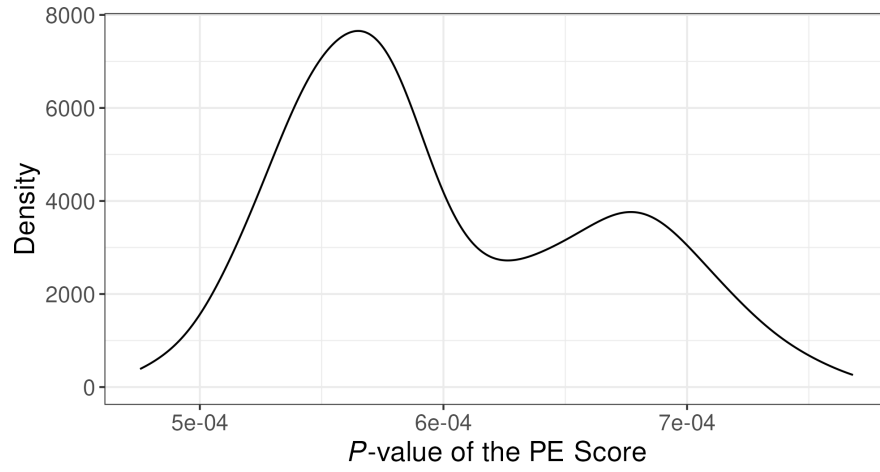


Note: This figure presents the distribution of the coefficient of the PE Score across the 1,000 bootstrap samples. The coefficient is obtained based on estimating the specification in column (6) of Table 5.

¹⁵As long as measurement errors in sales lift are uncorrelated with predictors of sales lift, the OLS coefficients of these predictors will still be estimated without bias.

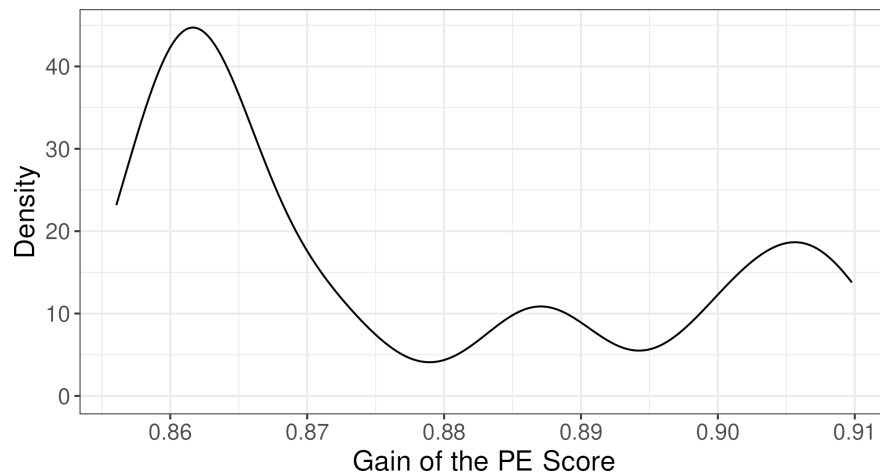
¹⁶We cannot similarly address the sampling variability of treated products because each DID regression in the first step is based on one treated product and 2,426 control products.

Figure H.2. Distribution of the P -Value of the PE Score Across Bootstrap Samples



Note: This figure presents the distribution of the p -value of the PE Score across the 1,000 bootstrap samples. The p -value is obtained based on estimating the specification in column (6) of Table 5.

Figure H.3. Distribution of the Gain of the PE Score Across Bootstrap Samples



Note: This figure presents the distribution of the gain of the PE Score in the XGBoost model across the 1,000 bootstrap samples.

H.5 Observed Engagement Measures as Predictors of Sales Lift

The main analysis in the paper uses each video's computed engagement score as a predictor of its sales lift. We check the robustness of the results if we replace a video's computed engagement score with observed engagement measures – the number of likes, comments, or shares.

Table H.7. Predicting Sales Lift
(Observed Engagement Measures as Predictors)

		Sales Lift		
		(1)	(2)	(3)
		Engagement as # Likes	Engagement as # Comments	Engagement as # Shares
Computed Scores & Observed Engagement	PE Score	100.62*** (28.84)	100.98*** (28.96)	100.81*** (28.80)
	Observed Engagement (in Millions)	24.76 (72.52)	603.40 (3,199.10)	732.34 (969.85)
	Product Score	−45.94 (29.30)	−47.63 (28.57)	−46.08 (27.83)
	Observed Engagement (in Millions) × Product Score	−107.71 (290.16)	−2,719.43 (8,948.66)	−5,366.47 (6,426.91)
Influencer Features	Gender	5.32 (6.41)	5.20 (6.36)	5.31 (6.35)
	# Followers (in Millions)	−0.02 (1.61)	−0.09 (2.13)	−0.62 (1.88)
	Average Play (in Millions)	0.22 (2.34)	0.32 (2.38)	0.63 (2.37)
	Price per Video Ad (in Millions)	−1.03 (204.01)	2.50 (193.85)	30.73 (193.00)
	Expected CPM (in Millions)	12.98 (43.97)	12.67 (43.97)	11.64 (43.92)
	# Video Ads Influencer Has Posted	−0.01 (0.18)	−0.01 (0.18)	−0.01 (0.18)
Product Features	Average Search	−3.35 (7.12)	−3.37 (7.12)	−3.65 (7.12)
	Price	0.003 (0.01)	0.003 (0.01)	0.002 (0.01)
	Discount	−0.002 (0.005)	−0.002 (0.005)	−0.002 (0.005)
	Product-Category Indicators	Yes	Yes	Yes
Observations		259	259	259
R ²		0.08	0.08	0.08
Adjusted R ²		−0.01	−0.01	−0.01

Note: The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level, or video-level, sales lift in 1,000 RMB. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table H.7 presents the OLS estimation results, where the PE score continues to be a positive predictor and the only significant predictor of sales lift ($p < 0.001$). We rerun XGBoost, replacing the computed engagement score with observed engagement. For a more succinct and strict test, we include all three measures of observed engagement (the number of likes, comments, and shares) simultaneously in the XGBoost model. (Including them one by one leads to similar results.) Table H.8 presents the most important features of the XGBoost model. The PE score continues to be the most important predictor by far of sales lift based on gain.

Table H.8. XGBoost Feature Importance in Predicting Sales Lift
(Observed Engagement Measures as Predictors)

Feature	Gain	Cover	Frequency
PE Score	0.876	0.338	0.242
Expected CPM	0.043	0.009	0.061
Product Score	0.036	0.328	0.182
Discount	0.017	0.034	0.212
# Comments	0.011	0.003	0.061
Average Search	0.008	0.142	0.061
# Shares	0.003	0.069	0.061
# Comments $\times$ Product Score	0.003	0.072	0.030
# Likes	0.003	0.001	0.030
Price	0.0002	0.002	0.030

Note: The sample consists of all treated products in the sales panel. The features are ranked by gain, the standard metric of feature importance for the XGBoost model's predictive accuracy. The table reports the 10 most important features of the XGBoost model trained on the entire sample.

H.6 Average Engagement and Product Scores as Predictors of Sales Lift

In the paper, a video’s engagement score is computed in the “sum form,” meaning as the *sum* of its pixel-level engagement values. A video’s product score is also computed in the sum form as the *sum* of pixels in which the product appears. Meanwhile, a video’s PE score is defined in the “average form” as the *average* of its pixel-level values (Online Appendix B). As explained in the paper, we use the sum form, rather than the average form, to compute the engagement and product scores to reduce their correlation with the PE score, while still capturing a video’s level of engagement and product placement.

Recall that a video’s pixel-level PE scores are the inner product of its pixel-level engagement scores and pixel-level product scores. The video-level PE score may thus be correlated with the average-form engagement and product scores. Intuitively, a video that is unengaging throughout or barely presents the product is likely to also have a low PE score. The degree of correlation is an empirical question. For products in the sales panel, the PE score has a marginally significant correlation ($\rho = 0.11$, $p = 0.09$) with the average-form engagement score, and has a high and significant correlation ($\rho = 0.85$, $p < 0.001$) with the average-form product score. This high correlation may obfuscate the analysis of sales lift predictors, in both the OLS and XGBoost frameworks.

Computing a video’s engagement and product scores in the sum form can mitigate this correlation, drawing on the variation in video length. (The sum of pixel-level values equals the average of pixel-level values multiplied by the number of pixels per frame, which is held constant in this paper, and multiplied by video length.) Indeed, when a video’s product score is computed in the sum form, its correlation with the PE score decreases to 0.54 ($p < 0.001$), which is still significant statistically but moderate in magnitude (Schober et al. 2018). The correlation between the engagement score and the PE score becomes insignificant ($\rho = -0.018$, $p = 0.77$) when the former is computed in the sum form. These sum-form product and engagement scores are what we use in the paper.

Nevertheless, we will test the robustness of the results when we compute a video’s

engagement score and product score in the average form, denoted as $\overline{\text{Engagement Score}}$ and $\overline{\text{Product Score}}$, respectively. For a closer look at these scores' predictive power, we redo the entire OLS Table 5, not just its column (6). Table H.9 shows the results. Furthermore, because video length is what distinguishes engagement and product scores in their average versus sum forms, we build on Table H.9 to control for video length in all of its specifications. Table H.10 shows the results.

The PE score is a positive and significant ($p < 0.01$) predictor of sales lift in both Tables H.9 and H.10, whether it is the sole predictor or joined by other predictors. The average-form engagement and product scores, as well as their interaction term, show less stable results; their predictive power depends on whether they are the sole predictors and whether we control for video length. These unstable results are common when predictors are highly correlated. The correlation concern is particularly visible in column (6) of Tables H.9 and H.10. Compared with their counterparts in column (6) of Table 5, the PE score coefficient is larger, whereas the product score coefficient becomes negative and significant. This change possibly reflects the high correlation between the PE score and the average-form product score, as discussed earlier.

Table H.11 presents the XGBoost feature importance results when the engagement score and the product score are computed in the average form. The top portion presents the results when video length is excluded from the feature list that enters the XGBoost model. The bottom portion presents the results when video length is included. The PE score again emerges as the most important sales lift predictor by far based on gain.

In summary, when the engagement score and product score are computed in the average form, the PE score remains a significant predictor of sales lift. The predictive power of the engagement score and product score are less stable, possibly due to their correlation with the PE score.

Table H.9. Predicting Sales Lift
(Average-Form Engagement and Product Scores as Predictors)

		Dependent Variable: Sales Lift					
		(1)	(2)	(3)	(4)	(5)	(6)
Computed Scores	PE Score	68.27** (22.87)				138.23** (44.79)	141.02** (46.86)
	Engagement Score		28.86* (13.52)			−19.91 (25.17)	−26.04 (26.48)
	Product Score			19.50 (16.14)		−106.84** (37.25)	−106.02** (38.57)
	Engagement Score × Product Score				88.92** (28.50)	125.54 (69.25)	127.62 (72.62)
Influencer Features	Gender						5.20 (6.28)
	# Followers (in Millions)						0.39 (1.46)
	Average Play (in Millions)						0.24 (2.30)
	Price per Video Ad (in Millions)						−19.34 (180.10)
	Expected CPM (in Millions)						10.68 (43.44)
	# Video Ads Influencer Has Posted						−0.01 (0.17)
Product Features	Average Search						−3.95 (7.02)
	Price						0.002 (0.01)
	Discount						−0.003 (0.005)
	Product-Category Indicators	No	No	No	No	No	Yes
	Observations	259	259	259	259	259	259
R ²		0.03	0.02	0.01	0.04	0.08	0.10
Adjusted R ²		0.03	0.01	0.002	0.03	0.06	0.01

Note: The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level, or video-level, sales lift in 1,000 RMB. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table H.10. Predicting Sales Lift
(Average-Form Engagement and Product Scores as Predictors,
Controlling for Video Length)

		Dependent Variable: Sales Lift					
		(1)	(2)	(3)	(4)	(5)	(6)
Computed Scores	PE Score	65.85** (23.00)				141.34** (44.96)	145.92** (47.12)
	Engagement Score		35.85 (20.42)			−8.23 (28.55)	−12.15 (29.88)
	Product Score			19.87 (16.12)		−109.99** (37.45)	−109.83** (38.75)
	Engagement Score × Product Score				94.14** (33.07)	129.04 (69.40)	131.23 (72.71)
	Video Length	−0.20 (0.20)	0.14 (0.30)	−0.26 (0.20)	0.07 (0.23)	0.25 (0.29)	0.31 (0.31)
Influencer Features	Gender						5.36 (6.28)
	# Followers (in Millions)						0.62 (1.48)
	Average Play (in Millions)						0.11 (2.31)
	Price per Video Ad (in Millions)						−44.36 (181.82)
	Expected CPM (in Millions)						12.84 (43.49)
	# Video Ads Influencer Has Posted						−0.01 (0.17)
Product Features	Average Search						−4.77 (7.07)
	Price						0.002 (0.01)
	Discount						−0.002 (0.005)
	Product-Category Indicators	No	No	No	No	No	Yes
	Observations	259	259	259	259	259	259
R ²		0.04	0.02	0.01	0.04	0.08	0.10
Adjusted R ²		0.03	0.01	0.005	0.03	0.06	0.01

Note: The sample consists of all treated products in the sales panel. Each treated product corresponds to one video ad. The specification is OLS. The dependent variable is the estimated product-level, or video-level, sales lift in 1,000 RMB. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

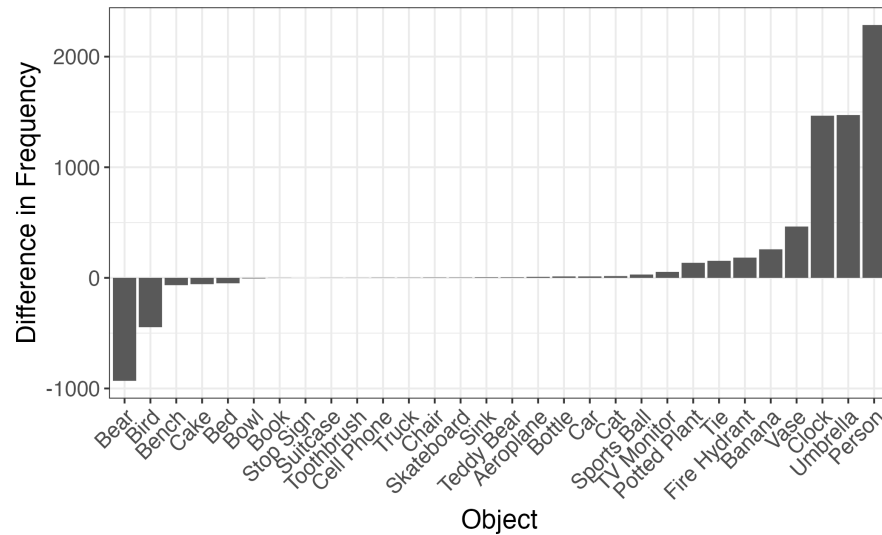
Table H.11. XGBoost Feature Importance in Predicting Sales Lift
(Average-Form Engagement and Product Scores as Predictors)

Feature	Gain	Cover	Frequency
Without Video Length as a Feature			
PE Score	0.869	0.344	0.235
Expected CPM	0.057	0.090	0.147
Engagement Score	0.040	0.400	0.353
Product Score	0.029	0.011	0.059
# Followers	0.002	0.077	0.118
Average Search	0.001	0.072	0.029
Discount	0.001	0.007	0.059
With Video Length as a Feature			
PE Score	0.869	0.344	0.235
Expected CPM	0.057	0.090	0.147
Engagement Score	0.039	0.330	0.324
Product Score	0.029	0.011	0.059
# Followers	0.002	0.077	0.118
Average Search	0.001	0.072	0.029
Video Length	0.001	0.070	0.029
Discount	0.001	0.007	0.059

Note: The sample for each portion of the table consists of all treated products in the sales panel. The features in each portion are ranked by gain, the standard metric of feature importance for the XGBoost model's predictive accuracy. Each portion of the table reports up to 10 of the most important features of the XGBoost model trained on its entire sample. When the number of features shown in a portion of the table is less than 10, the unshown features are not used by the XGBoost model to split nodes, meaning they are unimportant according to the model.

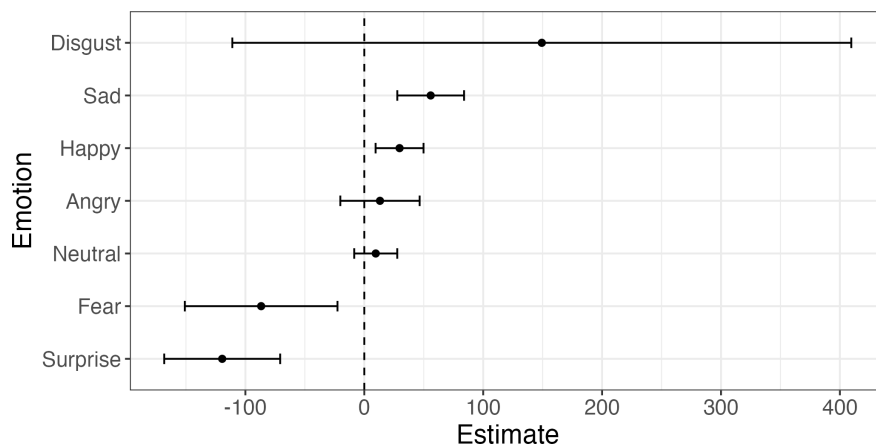
I Exploring Drivers of the PE Score

Figure I.1. Objects in High-Engagement versus Low-Engagement Pixels



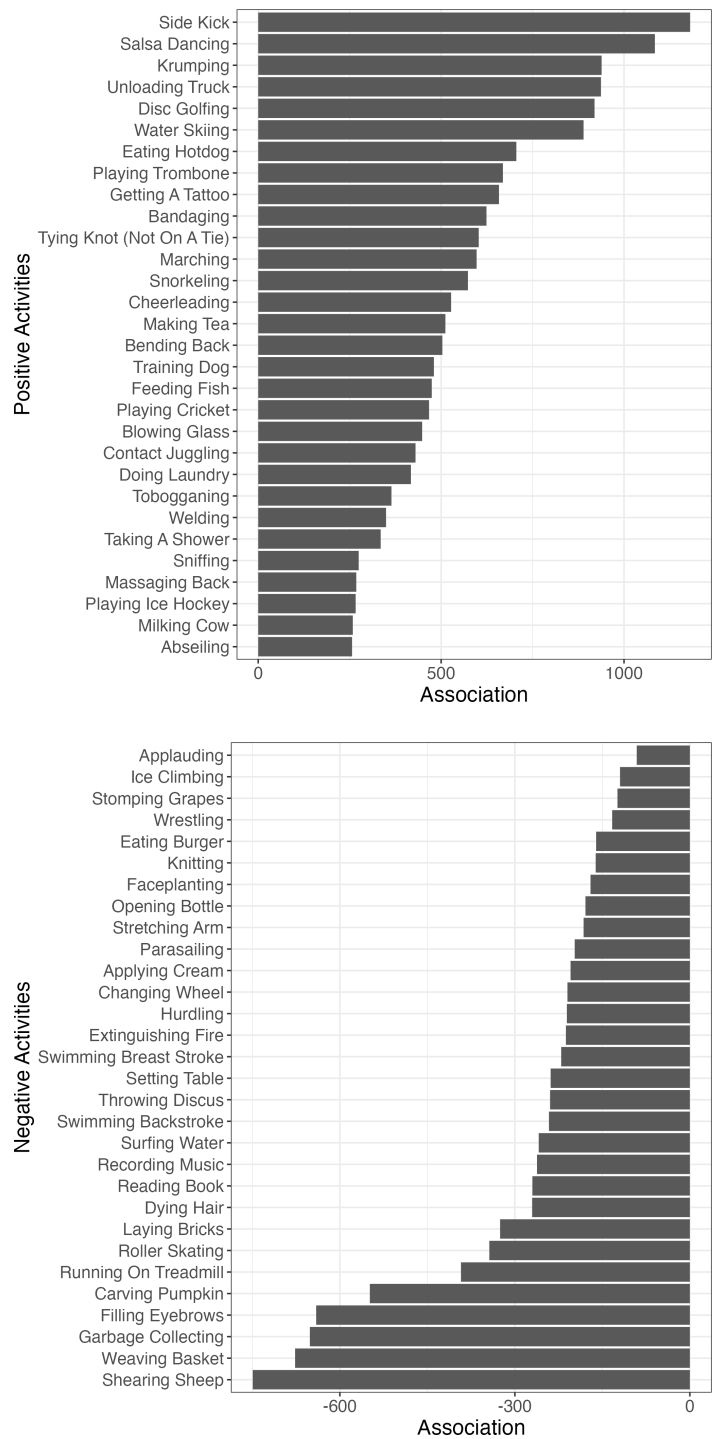
Note: The sample consists of all videos in the sales panel. Frequency is the number of times an object is detected in either the high or low version of videos in the sales panel. The difference in frequency is the frequency in high versions minus that in low versions. Some objects are only detected in high or low versions alone so that a difference is not computed, but these objects are rare.

Figure I.2. Emotions and Engagement



Note: The sample consists of all videos in the sales panel. Each observation is a frame of a video in the sales panel. The estimate is obtained by regressing the average pixel-level engagement scores of a frame on the probabilities of the detected emotions in that frame. Results are relative to a baseline where no emotion is detected. Bars denote the 95% confidence intervals.

Figure I.3. Association between Activities and Engagement Scores



Note: The sample consists of all videos in the sales panel. Each observation is a 15-second segment of a video in the sales panel. The association is obtained by regressing the average pixel-level engagement scores of a segment on the probabilities of the detected activities in that segment, controlling for segment sequence (e.g., the first in a video). Results are relative to a baseline where no activity is detected.

Table I.1. Activities and Engagement:
Top 10 Words in the Topic Model

Positive Activities		Negative Activities	
Topic 1	Topic 2	Topic 1	Topic 2
play	play	swim	appli
danc	make	paper	paper
make	clean	question	fill
clean	climb	tabl	swim
use	ice	fli	run
back	blow	feed	stretch
basketbal	car	die	leg
fold	eat	basket	wax
eat	floor	monopoli	facepl
push	head	bottl	book

Note: The sample consists of the detected activities positively or negatively associated with engagement based on all videos in the sales panel. The words presented in the table are in the format after stemming. For example, “dance,” “dancing,” and “dances” all become “danc” after stemming.

Figure I.4. Word Clouds from the Survey of Perceived Activity Characteristics



More-Engaging Activities

Less-Engaging Activities

Note: The sample consists of responses from a survey of 101 participants who tend to be familiar with TikTok. Each participant was asked to write three to five adjectives or phrases to indicate their perceived common characteristics of the more-engaging activities and the less-engaging activities, respectively.

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